BBE SCHOOL

Introduction to Business and Economics

Study material for the Economics Full Course
Chapters 2 – 6

Contents

2 Basic economic concepts

Before one can study how a business is organised and managed, it is necessary to understand the economic environment of which every business forms a part. Basic economic concepts shape the decisions of firms and of individuals: what to offer for sale, what to purchase, how to manage limited resources, and how to finance expenditure.

This chapter develops that shared foundation. It begins with participation in the economy and the condition of scarcity that forces every household, firm and government to choose. It then presents economics as the study of such decisions, and shows how exchange creates a circular flow of goods, services and money, together with division of labour. Different economic systems allocate decision rights between private agents and the state in different ways. Markets coordinate buyers and sellers through supply and demand, and competition among suppliers shapes the setting in which those markets operate.

2.1 Being part of the economy

By running a business, entrepreneurs become an active part of the economy. Businesses provide goods such as computers and services such as installing software for people who need these goods and services to satisfy their needs. Those needs might include having a computer as well as getting technical support. Not only individuals, but also businesses have needs: manufacturers of smartphones, tablets and laptops might need printed circuit boards produced by a specialised supplier. In turn, that supplier needs raw materials and a workforce to produce those boards. And individuals might also exchange goods and services with each other, like selling a house, an apartment or a used car, or exchanging vegetables or flowers from the garden and getting a bottle of wine in return. In the economy, people and businesses exchange goods and services to fulfil their needs and wants.

Therefore, people are part of the economy long before they start a business. As individuals, they belong to private households and buy goods and services from a wide range of businesses. Like thousands of other households, they need food, a home to live in, medical care, a car and/or other kinds of transportation; they want to do sports or go to the cinema, to a café or a restaurant. Businesses offer the goods and services that people - as part of private households or other businesses - need and/or want.

A business is an entity that offers goods and/or services to customers.

A business produces primarily for customers rather than only for the owners' private household needs, and usually charges a price so that something of value is received in return. The economy is the overall system of production, exchange and consumption through which households, businesses and public authorities interact to meet needs and wants.

Worked example. *A student works evenings in a café and spends wages on rent, food and a language course. She already takes part in the economy as a supplier of labour and as a consumer. If she later opens a tutoring service for secondary-school pupils, she remains a household member who buys transport and phones, but she additionally becomes an entrepreneur supplying education services. Both roles belong to the same economic system; only the mix of activities changes.*

- **Exchanging would be much easier if all that we want or need were available in abundance, but that is not the case.**
- **Resources are scarce and need to be managed.**
- **This is why we all need to economise. No one is able to opt out of making economic decisions.**

2.2 Scarcity of resources and opportunity cost

Both households and businesses only have limited resources to pursue their goals: businesses have a certain number of machines and tools and a limited amount of material and financial resources to produce their goods and provide their services. Likewise, households only have a certain income that they are able to spend on purchasing goods and services, with maybe a part of that income left for saving. Resources are always scarce, and this scarcity of resources forces businesses as well as individuals to make decisions on how to use those limited resources: what and how to produce, what to buy, how much to spend.

Scarcity is the basic condition that resources are limited while needs and wants are effectively unlimited, so that choices about allocation become necessary.

It is a basic economic problem of households, businesses and the government to decide how to use their limited resources and to make choices when allocating these scarce resources between different options. Two graduates can either found their own business or take jobs at another business and become employed. If they earned 35,000 euros per year in such a job, this 35,000 euros would be their opportunity cost per person if they decided to set up their own business. If they took these jobs, they would also have opportunity cost: the amount of money that they could have earned with their business. Of course, it would be difficult to tell how much that could have been.

Opportunity cost is the (financial) benefit of the (next best) alternative that is lost or given up in order to choose or achieve something else.

Opportunity cost makes trade-offs visible. Spending a Saturday revising accounting rather than working an extra shift has an opportunity cost in wages not earned, and possibly in leisure. Founding a firm instead of taking a salaried job means giving up that salary and its associated security - even if the new venture's accounts show no wage expense for the founder. Opportunity cost is often hardest to see when no cash changes hands: studying may look "free" of tuition fees at one institution, yet the next-best paid job still has a value that is given up.

Two clarifications matter. First, opportunity cost is the next-best alternative - the best option among those left behind - not the sum of every rejected dream. Second, scarcity is not the same as poverty. Even wealthy households and rich countries face scarcity because wants expand and resources remain finite for any given decision.

Worked example. *Two graduates can each earn €38,000 a year in employment. If they start a repair café instead, the opportunity cost of self-employment is at least that salary each - plus any benefits they would have received - because that package is the next-best realistic alternative. If the business later earns more than the combined forgone salaries after proper cost allowance, the choice can still be rational; the point is that zero reported wage for the founders does not mean zero cost of their time.*

2.3 Economics is the study of economic decisions

Economics is the study of how individuals (as part of private households) and businesses make decisions to satisfy their needs and wants with limited resources.

It comprises a number of scientific fields and branches, two of which are microeconomics and macroeconomics. Microeconomics focuses on the behaviour and decisions of individual households and businesses and how they interact. A question in microeconomics could focus on the change in demand for electric cars if buyers of electric cars receive a bonus. In contrast, macroeconomics looks at the bigger picture and deals with questions concerning the overall economy (of one country for example) and aggregate quantities. Among other phenomena, it studies economic growth, unemployment, interest rates, price levels and inflation.

As a science, economics strives to explain the observed phenomena and also to make predictions, both based on various theories. Models deliberately omit detail so that cause and effect become visible. Predictions are conditional: if incomes rise and tastes stay constant, demand for normal goods tends to rise. Understanding the "if" is as important as remembering the conclusion.

The two lenses are complementary. A wage rise in one firm is a microeconomic issue; rising unemployment nationwide is macroeconomic. Yet firm-level hiring accumulates into national employment figures, and national interest-rate policy feeds back into household borrowing and firm investment. Economic decisions appear everywhere: whether to hire or automate, rent or buy, specialise or diversify, tax or borrow, consume now or save. Business studies then ask how organisations design products, finance assets and manage people inside that environment.

Worked example. *Question A asks how a €2,000 purchase bonus for electric bikes would affect demand for e-bikes in one city. That is microeconomics - buyers and sellers in one product market. Question B asks how inflation and unemployment evolved in the euro area last year. That is macroeconomics - aggregate indicators for a large economic area.*

2.4 Exchanging goods and services creates a circular flow and division of labour

While households mainly offer labour and receive wages, businesses offer goods and services that are bought by households and other businesses and receive money for what they sell. This is how a circular flow of goods, services and money is created. As money is used as a means of exchange that is widely used and accepted, these exchanges can be carried out relatively easily. Without money, people would have to barter, which is much more complicated. If you want to acquire a good from another person, but do not have what this person wants in return, there will be no exchange.

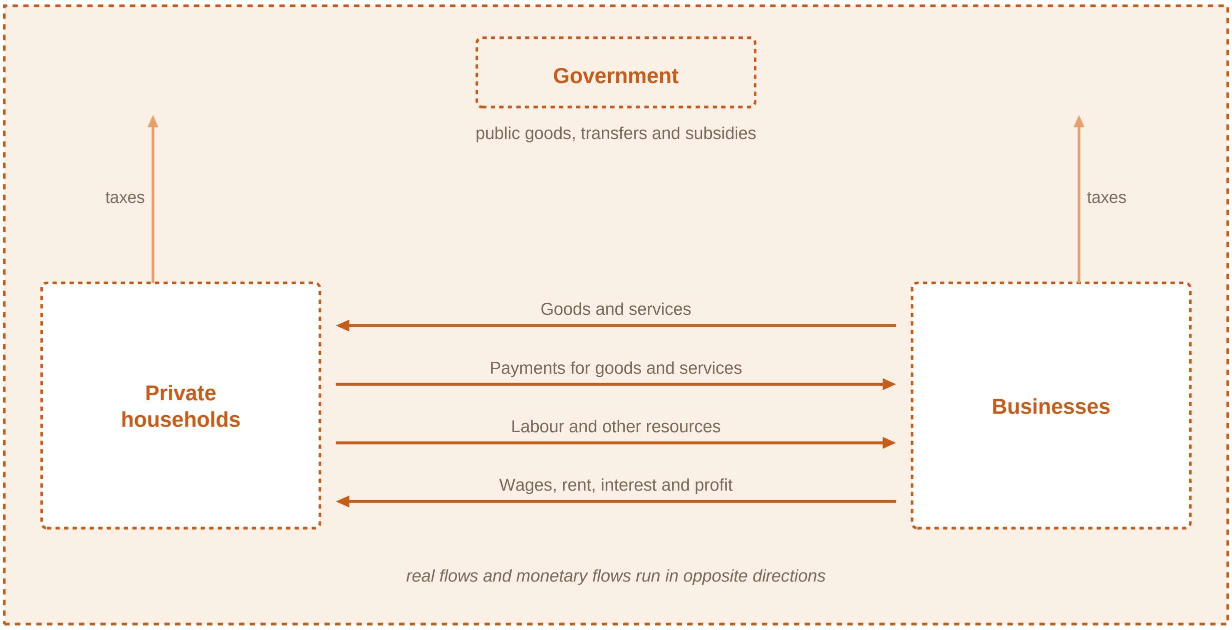

Figure 1. The circular flow of goods, services and money within the economy - households, businesses and government linked by labour, wages, products, payments, taxes and public services.

The circular flow describes the continuous movement of resources, goods and services in one direction and monetary payments in the opposite direction between households, businesses and government.

Money allows for a flexibility of exchange (first function: medium of exchange). It also allows us to express the value of things (second function: unit of account) and to store value over time (third function: store of value). Money fulfils these functions best if its value remains pretty stable over time. However, if there is a general rise in prices of goods and services, you can only buy a lower amount of goods and services, which means that the purchasing power of money (i.e. its value) declines.

- **Medium of exchange: money allows flexible trade without barter.**
- **Unit of account: money expresses and compares the value of things.**
- **Store of value: money can carry purchasing power into the future when its value remains reasonably stable.**

Price indexes allow us to measure the extent of this general increase in prices, a phenomenon that is also called inflation. Low inflation rates can be tolerated. The European Central Bank considers an inflation rate of slightly below 2% per year most beneficial to the economy. But if inflation rates are considerably higher, the purchasing power of money decreases considerably. As the amount of goods and services that can be bought for a certain amount of money continuously decreases, people lose their trust in money and try to get rid of it.

Inflation is a general rise in the prices of goods and services, which reduces the purchasing power of money.

In the circular flow of the economy, public authorities (mainly governments) also play an important role. They levy taxes from households and businesses and use the money to provide goods, transfer payments and subsidies. Such goods as infrastructure (e.g. streets and street lights) and services like national defence and public security (e.g. police) need to be provided by governments and financed by taxation. There is demand for these goods, but private businesses would not want to supply them as "free riders" (people who do not pay for a good or service) cannot be excluded from enjoying them. In many - but not all - countries, health care and education are also provided (at least to a large extent) by public authorities.

Without exchange of goods and services, individuals would have to produce everything they need themselves. That would be very difficult, time-consuming and inefficient because they would have to spend time on performing tasks that they lack the skills for. Exchanging goods and services allows for the division of labour and therefore specialisation. Individuals and businesses can concentrate on what they can do best. This explains the wide variety of jobs and of businesses. Using a widely accepted means of exchange like money facilitates the exchanges.

Specialisation can be found on many levels. Within households, for example, individuals can concentrate on what they can do best or what they like doing (like one individual does the shopping, the other one the cooking). The same principle applies to businesses: within businesses, some people concentrate on production, others on procurement, some others on sales, on recording all financial transactions or on managing human resources. Accordingly, these different tasks of a business are often summarised in different departments: procurement, production, sales, marketing, finance and accounting.

Specialisation can also be found between businesses as every business focuses on a special range of products: some offer all kinds of furniture, others just beds and couches, some others just produce kitchens. These businesses operate on the same level of production. But specialisation can also be based on division of labour between businesses on different levels of production (first level: production of wood and iron, second level: production of boards and nails, third level: production of tables, final level: selling them) or in different sectors of the economy (see Chapter

3).

Specialisation can also be found on an international level, as countries differ in climate, natural resources, geographical position and many other characteristics. Due to very different characteristics, countries also differ in their conditions for different industries (industry being a number of businesses that produce or sell the same product) and different business functions. A printed-circuit-board manufacturer, for example, may have production sites in Europe as well as in Asia. While production in Europe may be highly diversified and relatively low in volume, production in Asia may reach a much higher volume but have a lower product diversity. This kind of division of labour can be explained by differences in terms of know-how of the workforce, labour costs, availability of other resources and the legal framework in different countries.

While division of labour has many advantages, it also has some disadvantages that need to be considered: for very specialised workers, work may become boring over time. Being specialised also means less flexibility as it is hard to develop other skills or develop competencies in other fields. A specialised business may be brilliant in that field, but if - for some reason - that specialisation is not needed anymore, the business is at risk and people could lose their jobs.

Worked example. *A bicycle repair shop specialises in assembly and aftercare; a separate supplier makes frames; a courier firm delivers parts. Money lets each focus: the shop need not barter repaired bikes for tubing. If inflation jumps sharply, the shop's cash balances buy fewer spare parts over a few weeks, showing that the store-of-value function weakens when the general price level rises quickly.*

- **Specialisation appears within households, within firms, between firms, and internationally.**
- **Money's three functions facilitate exchange; they are not themselves a guarantee of wealth.**
- **Division of labour raises productivity but can make work monotonous and reduce flexibility.**

2.5 Different economic systems

The role of governments described above is mainly true for the governments in some sort of market economy. While in market economies individuals and businesses are - more or less - allowed to make many of their own economic decisions, in planned economic systems the governments play a dominant role. They (mainly or partly) decide which goods are produced and which services are offered (at which prices). Furthermore, they (mainly or partly) control the resources and the means of production. People have a limited choice of which job to do and which products to buy.

Market economy In a market economy, individuals and businesses make many of their own economic decisions about production, consumption and prices, within a legal framework.

Planned economy In a planned (or centrally planned) system, government mainly or partly decides which goods and services are produced, at which prices, and controls key resources and means of production.

In most countries of the world, economies can be characterised as some kind of market economy. In some of these countries, the governments play a minor role by only providing the legal framework and not influencing the economy much, so their economic system comes close to what is called a "free market economy". In many other countries, the government plays a more important role by influencing the economy to a somewhat higher extent, by supporting the poor and protecting the environment for example. These systems are called "social market economies" or "eco-social market economies".

Feature	Free market economy	Social market economy	Planned economy
Who mainly decides what is produced?	Private households and firms through markets	Private agents through markets; state sets social and environmental frameworks	Government (mainly or partly) through plans and quotas
How are prices determined?	Mostly by supply and demand	Mostly by supply and demand, with selected regulation and taxes	Often set or tightly controlled by the state
Role of government	Legal framework; comparatively light economic intervention	Legal framework plus stronger social protection, redistribution and environmental policy	Dominant control of production, resources and often employment
Ownership of means of production	Predominantly private	Predominantly private, with public enterprises in selected areas	Predominantly public or tightly state-controlled
Consumer and job choice	Wide	Wide, within legal and social rules	More limited

Table 1. Comparison of free market, social market and planned economic systems

Over the past decades, a lot of former communist countries that used to have planned economic systems have adopted the main principles of the market economy. Many Central and Eastern European Countries (CEE), China and former Soviet countries are examples of such a transformation. Transformation is rarely overnight: legal institutions, property rights, competition culture and safety nets must catch up with price liberalisation.

Worked example. *Country A lets private firms choose products and set most prices; consumers choose freely among suppliers; government mainly enforces contracts and competition law. That leans towards a free market pattern. Country B also uses private ownership and prices, but finances universal healthcare, generous unemployment support and tight environmental standards through taxes. That is closer to a social market pattern. Country C issues production quotas for steel and bread, sets official prices, and assigns workers to plants: that matches planned-economy features.*

2.6 Supply and demand: households, businesses and the government meet in the market

In a market economy, goods and services are offered and sold on markets. Buyers and sellers meet to communicate the conditions of exchanging goods and services and thus form a market. A market can be an actual place, like a flower market in the city of Rome or a flea market in a small village, but it can also be a virtual place like a market on the Internet. Buying and selling take place in shops, but also on the phone. Therefore, there are many different markets, also depending on what is offered: consumer goods markets, labour markets (supply and demand for work), housing markets, money markets, capital markets, commodity markets (supply and demand for raw materials).

A market is the interaction of buyers and sellers who communicate and agree, implicitly or explicitly, on the terms of exchanging a good or service.

Supply of a certain good or service is the quantity of that good or service that is available for purchase. Basically, this quantity mainly depends on the businesses' production capacities and the available resources, but also on the price that can be charged for the good or the service. The higher this price is, the higher the supply will be. All other things held constant ("ceteris paribus"), this relationship is true for most goods and services in the economy (law of supply).

Law of supply Ceteris paribus, a higher price leads to a higher quantity supplied for most goods and services.

In a tutoring market, for example, the quantity of hours that tutors are willing to work depends on the price they can charge and that will be paid. It is skilled work. If the price is low, say below 30 euros per hour, few providers would offer that service. The opportunity cost would be too high: as the providers are skilled, they could earn more money by doing something else, so they would leave this market. The higher the price, the higher the number of hours that would be offered. More potential providers would enter the market and would be willing to offer a higher number of hours of their service.

The shape of the supply curve can also be explained by the concept of increasing marginal costs faced by many industries and businesses. Marginal cost is the cost of producing an additional unit of a good or by providing an additional unit of a service. As output increases and exceeds a certain level it will become more and more costly to produce; businesses need to build higher capacities (machinery, personnel) so marginal costs would rise. Only if the price level increases and exceeds (or at least equals) marginal costs would businesses be willing to produce and supply a higher amount.

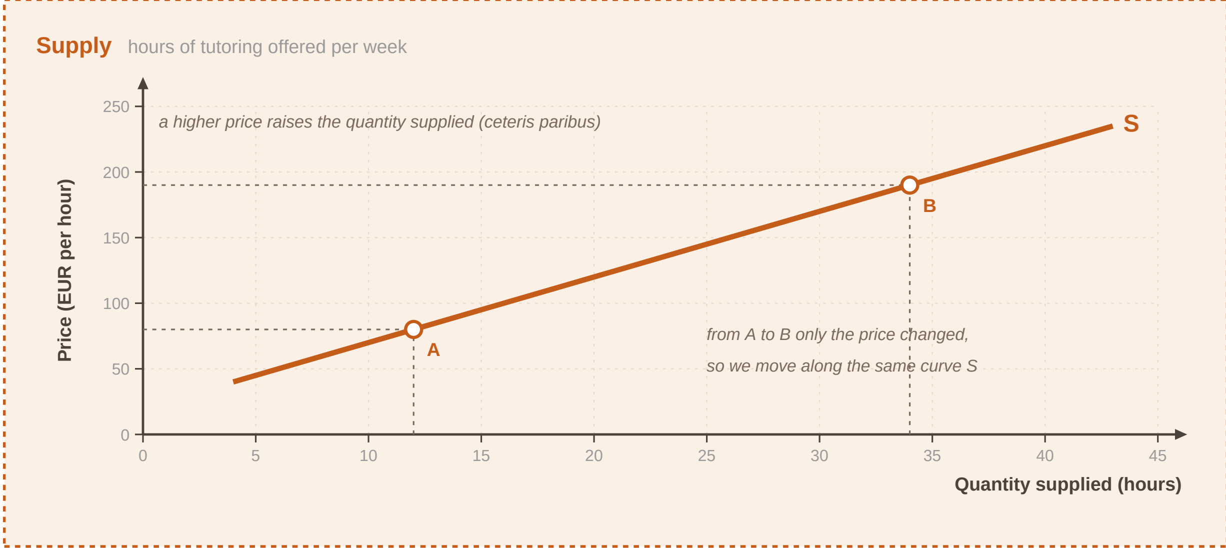

Figure 2. The supply curve - price on the vertical axis, quantity on the horizontal axis; the curve slopes upward as quantity supplied rises with price (ceteris paribus).

Supply relationship (ceteris paribus): As P rises -> Qs rises (law of supply). Movement along the supply curve: a price change alone changes quantity supplied. Shift of the supply curve: a non-price factor changes supply at every price.

Demand, on the other hand, is the quantity of a good or service that customers are willing and able to buy. Usually the higher the price is, the lower demand will be. The more tutors charge for their support service per hour, the more demand will decrease (because more and more people cannot afford such high prices or are not willing to pay such high prices for that service and will instead look for other ways to get help). People's willingness to pay a certain price is related to the utility or the level of satisfaction the people get from consuming the service.

Law of demand Ceteris paribus, a higher price leads to a lower quantity demanded for almost all goods and services.

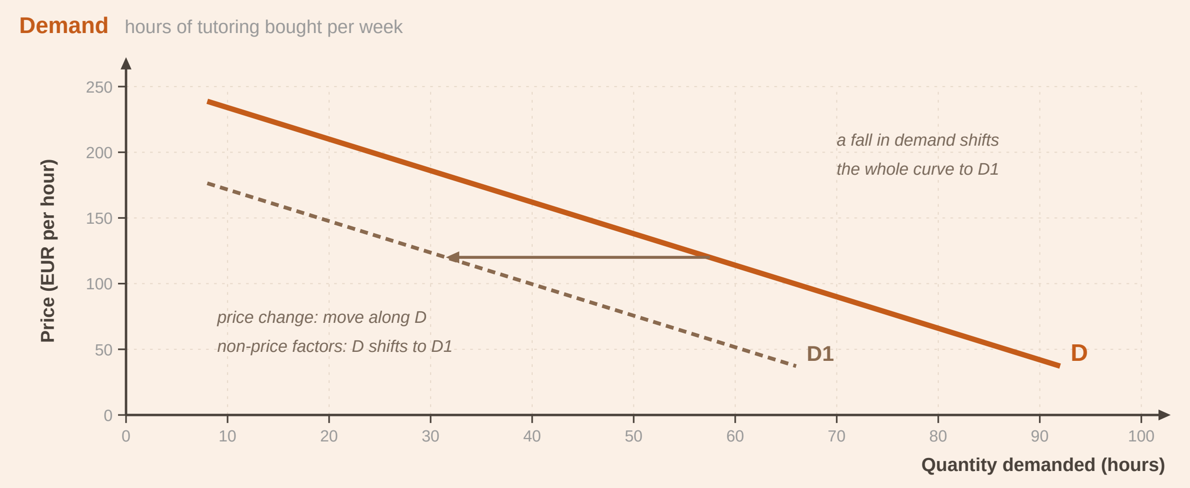

Figure 3. The demand curve - quantity demanded falls as price rises (ceteris paribus), so the curve slopes downward.

Demand relationship (ceteris paribus): As P rises -> Qd falls (law of demand). Movement along the demand curve: price change alone changes quantity demanded. Shift of the demand curve: income, tastes, complements, substitutes or other non-price factors change demand at every price.

If we have a look at both curves, we can see that they intersect at a certain point. At this point, the quantity of hours that is demanded in the market equals the quantity of hours that is supplied. Assuming that these curves represent supply and demand in the whole market, this price would also be called the market price or equilibrium price (because supply equals demand). At that price, demand and supply balance each other out. As the quantity supplied equals the quantity demanded, there is neither a surplus (higher supply than demand) nor a shortage (higher demand than supply). At a higher price, demand would be lower than supply and vice versa.

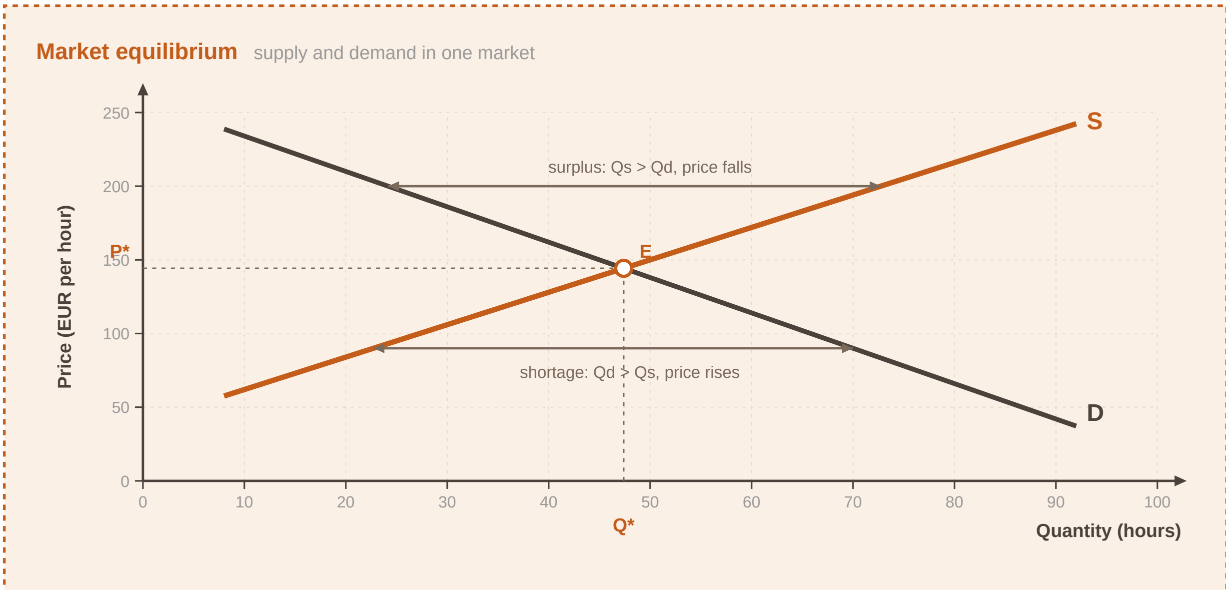

Figure 4. Market equilibrium - supply and demand intersect at the market (equilibrium) price, where quantity supplied equals quantity demanded.

Equilibrium condition: At equilibrium: Qs = Qd = Q*. If P > P*, typically Qs > Qd (surplus) and pressure for price to fall. If P < P*, typically Qd > Qs (shortage) and pressure for price to rise.

In the real world, other factors than price also affect demand and supply.

- **Demand shifts with changes in income, consumer preferences, complementary goods and the availability of substitutes.**
- **Supply shifts with the number of suppliers, technological changes, resource prices and price expectations.**
- **A change in price alone moves quantity along a curve; non-price factors shift the whole curve.**

The laws of supply and demand also help to understand one of the main causes of inflation. If the quantity of money within a country is increased (in order to stimulate the economy), people and businesses are able to buy more and to invest, so demand for goods and services usually rises. If the quantity of available goods and services in this country remains the same and does not increase accordingly, then the prices for goods and services will rise. "Too many dollars chasing too few goods" expresses very well what inflation is about. It is by increasing the price for money, i.e. the interest rates, that inflation can be fought. If it is more expensive to borrow money, people and businesses tend to spend less, demand for goods and services decreases again and so do their prices.

Worked example. *Suppose tutoring equilibrium is €40 per hour with 1,000 hours traded each week. If incomes rise and more families can afford tutoring, the demand curve shifts right: at the old €40 there is now a shortage, so the new equilibrium price and quantity are both higher under a standard upward-sloping supply. If software suddenly makes tutors more productive and lowers their costs, supply shifts right: equilibrium quantity rises and equilibrium price tends to fall.*

2.7 Competition in the market

The level of competition in a market is mainly influenced by the number of suppliers and the availability of substitute goods. If there is just one supplier, the market situation is called a monopoly. Monopolies are rare in a market economy but there might be goods and services that are only provided by one business (like the railway in some countries).

Some businesses might not be the sole suppliers in the whole market, but they might be the only supplier within a certain area, which also makes them a kind of monopolist. The owners of a ski hut might be in a monopoly-like situation if they are the only ones to offer food and beverages on a particular mountain. A theatre bar is in a similar situation. A tutoring service could find itself in such a situation if it is the only one that offers such a service within a radius of 50 km, for example.

A monopoly exists when there is only one supplier in a market.

If there are a few suppliers, the market form is called an oligopoly. Each supplier has a relatively large share of the market (e.g. telecom companies, car manufacturers). Competition can be strong, because as soon as one supplier changes the product or the price, its competitors are very likely to react in order to maintain their share of the market. Alternatively, suppliers could try to negotiate their terms of sale in order to prevent such harmful competition. Such agreements - also called a cartel - are usually not considered legal. In general, laws support competition in a market because it is considered beneficial for customers.

An oligopoly is a market with few suppliers, each holding a relatively large share.

Perfect competition can (theoretically) be found in markets with so many suppliers and buyers that no single individual or business can possibly influence the price. Considering perfect competition, in economics we also assume the following (theoretical) prerequisites:

- **All market players (buyers and sellers) must have access to all information at all times.**
- **There must not be any barriers to enter or exit the market.**
- **There must not be any (personal) preferences - goods must be replaceable (homogeneous).**

In real life, perfect competition is rarely found but, nevertheless, some markets come close to the concept of perfect competition. These are markets for goods and/or services that are almost identical (regardless of the supplier), e.g. agricultural markets, or that can be standardised. Even if the number of suppliers is not so high, competition can be almost "perfect" if competition among suppliers is especially fierce.

Worked example. *A remote valley has one grocery van that visits twice a week and no other store within an hour's travel - locally monopoly-like for many staples. A national mobile phone market with three large network operators is oligopolistic: a promotional tariff by one usually prompts matching offers. A large wholesale market for standardised wheat grades, with many growers and buyers, comes closer to perfect competition than either example.*

3 Focus on different types of businesses

Private households come in many different forms - large families, singles, couples, with or without children - and so do businesses. The large variety of businesses is due to the fact that they differ in the factors of production that they combine, in the sector(s) they operate in and in size. But they all need to consider their stakeholders around them and the environment.

This chapter trains the reader to classify businesses along those dimensions. The same enterprise can be described as tertiary, profit-oriented, micro and local, and stakeholder mapping completes the picture.

3.1 Businesses combine different factors of production

As mentioned earlier, a business is an entity that offers goods and/or services to customers. In order to do so, it combines different factors of production, the resources used to create goods and services. The most important factors are labour (all human resources), land (all natural resources), capital (resources like machinery, plant, vehicles, financial resources), and entrepreneurship (which brings land, labour and capital together) as well as knowledge and technology.

Factors of production are the resources used to create goods and services: labour, land, capital, entrepreneurship, and knowledge and technology.

Depending on the factors that are dominant for the production process, a wide range of businesses can be found. A tutoring start-up mainly combines knowledge and skills, labour, technology and some capital as well as entrepreneurship. A printed-circuit-board manufacturer combines all factors of production. A winemaker with large vineyards combines land, labour, capital and entrepreneurship, but the business might also be dependent on knowledge, experience and some technology.

Worked example. *A hillside orchard uses land (orchard plots), labour (seasonal pickers and permanent caretakers), capital (tractors, storage and sorting equipment, working capital), entrepreneurship (owners who decide crops and carry sales risk), plus knowledge (pruning methods, pest management) and technology (irrigation controls). A city language school in rented rooms uses little agricultural land, but still uses location as a resource, relies heavily on teacher labour and curriculum knowledge, needs capital for rooms and platforms, and depends on founders' entrepreneurship to attract students.*

- **Labour covers all human resources applied in production.**
- **Land means natural resources used in production.**
- **Capital includes machinery, plant, vehicles and financial resources.**
- **Entrepreneurship brings the other factors together and bears business risk.**
- **Knowledge and technology reshape how factors combine.**

3.2 Businesses operate in the primary, secondary and/or tertiary sector

Depending on what a business does, it contributes to one (or more) of three sectors of the economy. The three-sector model differentiates between three sectors of activity.

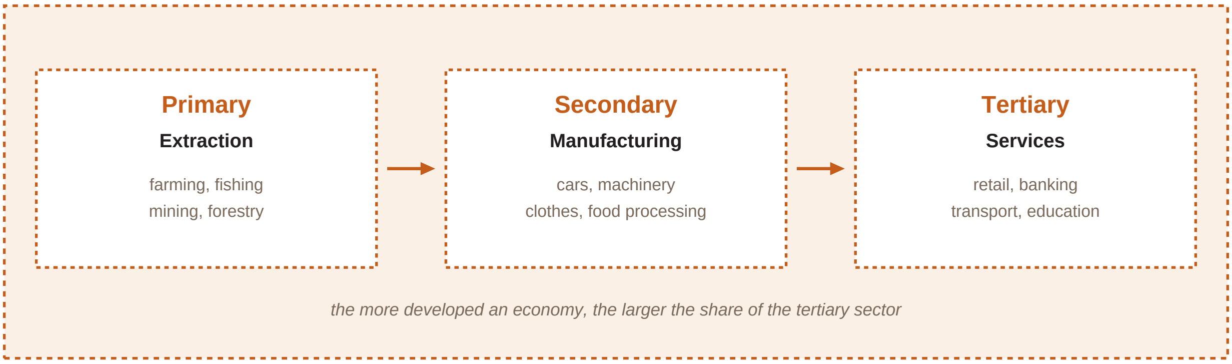

Figure 5. The three economic sectors - primary (extraction of raw materials), secondary (manufacturing) and tertiary (services), which together structure economic activity.

The primary sector refers to the extraction of raw materials from the earth. It mainly comprises farming, fishing, mining and forestry. Emerging countries (which are economically less developed) usually depend largely on the primary sector. Businesses of the secondary sector transform raw materials into goods (manufacturing). Such businesses produce cars, ships, machinery, printed circuit boards, computers, clothes and so on. The tertiary sector comprises the service industry, like distribution, banking, insurance, coaching and technical support.

The three-sector model classifies economic activity into primary (extraction), secondary (manufacturing) and tertiary (services).

The more the economic development of a country advances, the more important the third sector becomes. In economically highly developed countries, like EU countries with a high gross domestic product (GDP) per capita (indicating a high standard of living), the tertiary sector usually accounts for more than 70% of the output of the economy, while the primary sector - as important as it may be for providing food - only accounts for a small percentage of the output of the economy.

Gross domestic product (GDP) GDP is the total monetary value of final goods and services that are produced within a country's borders in a certain time period (usually a year).

Therefore, GDP is considered to measure the overall economic activity of a country. GDP is also used as an indicator for economic growth: if GDP (adjusted for inflation) increases over time, the economy is growing. However, the use of GDP is not undisputed. Critics say that not all sources of income are taken into account. What is more, GDP does not tell anything about the quality and/or sustainability of growth, nor is it a perfect indicator of economic well-being in a country. For example, GDP would also increase after ecological disasters that require action to rebuild infrastructure and to mitigate the damage. However, GDP is usually correlated to some indicators of well-being like health status and happiness.

Worked example. *A mining company extracting copper operates in the primary sector. A plant turning copper into cables is secondary. A utility that installs and maintains grid connections for households is tertiary. A retailer selling packaged food is tertiary even though food originated in farming; the retailer's main activity is distribution and customer service, not extraction.*

3.3 Businesses can be profit-oriented or not-for-profit

Most businesses aim to operate for a longer time period. In order to do so and to thrive over time, most businesses aim to make a profit, i.e. they intend to have higher revenues than costs and expenses. Profits are important to the business itself because they can be reinvested in the business, which enhances the durability and sustainability of the business. Of course, profits are also important to the owners and investors because the profits are their reward for the risk they have taken.

However, there are also businesses (or organisations) that are not-for-profit and mainly aim to cover their costs. Nevertheless, they also need to achieve a certain level of revenues (or in many cases: donations) in order to be able to offer their goods and services. Any profit they make is also beneficial to the organisation, because it can be reinvested and used to enhance the services for customers or to engage in more projects. The Red Cross, the World Wildlife Fund or Greenpeace are examples of such non-profit organisations (NPOs).

Aspect	Profit-oriented business	Not-for-profit organisation (NPO)
Primary aim	Earn profits (revenues above costs) as a central goal	Cover costs while pursuing a mission
Use of surplus	Reward owners/investors for risk; reinvest to strengthen the firm	Reinvest to enhance services or projects; not distributed as private profit goal
Need for revenue	Sales and other commercial income required	Donations, fees, grants or membership income still required
Success focus	Financial return alongside viability and competitiveness	Mission delivery and financial sustainability
Typical examples	Private firms, commercial partnerships, profit-seeking corporations	Charities, humanitarian NGOs, many cultural and environmental associations

Table 2. Profit-oriented businesses and not-for-profit organisations compared

A profit-oriented business aims to earn profits - revenues higher than costs and expenses - so that owners are rewarded for risk and the firm can reinvest.

Worked example. *A private language school that expands only when expected profits justify investment is profit-oriented. An international relief organisation that raises donations to provide emergency aid, covering overheads and reinvesting any excess into more programmes, is not-for-profit. Both must manage cash carefully; only the first treats owner profit as a core success metric.*

3.4 Businesses come "in all sizes": large and small

There is an enormous variety of businesses: micro, small and medium enterprises (MSME or just SME) as well as large businesses. About 99% of all businesses in the EU are SMEs. The European Commission differentiates between these types of businesses as presented in the table below.

Company category	Staff headcount	Turnover	OR Balance sheet total
Micro	< 10	≤ €2 m	≤ €2 m
Small	< 50	≤ €10 m	≤ €10 m
Medium-sized	< 250	≤ €50 m	≤ €43 m
Large	Above SME ceilings	Above SME ceilings	Above SME ceilings

Table 3. EU size categories of businesses (SME definition)

SME is the umbrella for small and medium-sized enterprises below large-enterprise thresholds, further distinguished as micro, small and medium-sized.

The definition of an SME is important for access to finance and EU support programmes targeted specifically at these enterprises. A two-person repair service that hires two assistants remains a micro company. A manufacturer with about 10,000 employees, on the other hand, is a large company.

The size of a business is also relevant in terms of legal requirements for their accounting (see Chapter 6). In many countries, small businesses are allowed to use simpler accounting methods in order to calculate their profits or losses.

Worked example. *Studio A has 6 employees and €1.2 million turnover: micro. Studio B has 40 employees and €8 million turnover: small. Factory C has 180 employees, €40 million turnover and a €30 million balance-sheet total: medium-sized. Group D has 8,000 employees worldwide: large. Eligibility for an SME innovation grant would typically open to A-C, subject to full legal tests, and exclude D.*

3.5 Businesses may be local, national or international

A local (or regional) business operates in a small, limited area. Most customers live very close to the business. They do not serve a national or international market. A neighbourhood bakery or a new tutoring service would be such a local business, at least at the beginning. The most important challenge of (small) local businesses is to have sufficient financial funds and to acquire a substantial number of customers. Many business owners face the difficulties of undercapitalisation: their own funds are limited and it is very hard for them to access additional funding.

A national business serves the home market (the market within a country), but not the international market. An insurance company or a bakery that operates all over Austria, but not abroad, is a national business. Like a local business, it has to choose a suitable location. Additionally, it has to decide how to deliver its goods and services to other places within the country. The supply chain is much longer.

International (or multinational) businesses make and/or sell their goods and services in more than one country. As the home market might be limited and too small for some companies, they strive to serve a larger market. A larger, international market brings along a number of challenges: an even longer supply chain, different legal and economic systems, different cultures and languages, maybe also different currencies. As the number of multinational businesses has increased over time, the world has become globalised and we have experienced globalisation. A components manufacturer with production plants in Asia as well as in Europe that sells its goods to other businesses all over the world is an international business.

Globalisation refers to the deepening cross-border integration of markets, production and information as more firms operate internationally.

Worked example. *A neighbourhood bike workshop whose clients all live within a few kilometres is local. A supermarket chain with stores in every major city of one country but none abroad is national. A components manufacturer with plants in Europe and Asia that sells to industrial clients worldwide is international. Each faces a distinct logistics and regulatory map.*

- **Local businesses serve a limited area and often face undercapitalisation.**
- **National businesses serve the home-country market but not foreign markets.**
- **International businesses make and/or sell in more than one country.**

3.6 Businesses operate in an environment - stakeholders are important

There is one thing that all businesses have in common: they are always integrated into an environment in which they operate. Therefore, they deal with many other people and/or businesses and need to consider their interests as well. Everyone that is (potentially) affected by the activities of a business and/or has an interest in what a business does is a stakeholder. Due to globalisation, the world has become a smaller place and the number of stakeholders has grown considerably. Furthermore, people are more environmentally conscious and expect businesses to preserve and protect the environment.

A stakeholder is anyone who is (potentially) affected by the activities of a business and/or has an interest in what a business does.

Stakeholder group	Typical interest / concern
Owners / investors	Return on capital for risk taken; value growth; sometimes mission and social contribution
Managers	Income, career identity, ability to implement strategy; may or may not also be owners
Employees	Wages, job security, meaningful work; shared values with the organisation
Customers	Useful, reliable offers at acceptable prices; ongoing relationship
Suppliers	Timely orders, fair payment, stable future demand
Government	Tax compliance, observance of rules, contribution to public aims
Communities (local, national, international)	Jobs, congestion, pollution, cultural sponsorship, licence to operate
Natural environment	Sustainable use of resources; reduced emissions, waste and ecological harm

Table 4. Main stakeholder groups of a business and typical interests

The owners of a business are stakeholders because they have invested their money in the business and want their investment to pay off. They want to make a profit in order to get something in return for the risk they have taken. They also profit from the increase in value of a business if the business operates successfully on the market. This increase is usually reflected in the price of shares of this business. It can be earned by selling the shares or the whole business. Making a profit may be one, but does not have to be the only goal of the owner(s) of a business. Solving a problem, contributing to a solution to a problem that customers have, and making a contribution to the welfare of society may also be important.

Managers (who can, but do not necessarily have to be owners) as well as employees are stakeholders because they depend on the business. Working for the business is their source of income, so job security is important. What is more, many managers and employees identify themselves with their tasks and their contribution to the business. And just as they depend on the business, the business depends on them as well. Only if owners, managers and employees share the values and objectives can a business succeed not only in the short run, but also in the long run.

Suppliers are important to a business and its production. Timely deliveries of good quality and the correct quantity of supplies needed for production are crucial. If a supplier fails, a business cannot produce its goods or services. In return, suppliers expect to get paid and hope for more orders in the future. If a business fails, suppliers may also get into trouble. A similar mutual dependency exists between a business and its customers.

Local, national and international communities as well as the government also depend on a business and are affected by its activities. Finally - and most importantly - businesses also need to act responsibly with the environment and the natural resources that they use. It is not sufficient only to claim to be environmentally friendly and to care about the environment (to "green wash" the business activities), but it takes concrete activities and proven results. More and more businesses report their activities on their websites and in their annual reports.

There are many factors that influence the success of a business. Considering the stakeholders and their (conflicting) interests is one of these factors. Other important factors include a suitable legal structure and a solid financial structure of the business, awareness of the market and changes in the market, awareness of costs and profitability. The following chapters deal with these topics.

Worked example. *A mid-sized food processor lists stakeholders: family owners (dividends and continuity), plant managers, production staff, farmers who supply milk, supermarket buyers, the municipal government (permits and taxes), nearby residents (noise and traffic), and the river ecosystem affected by wastewater. A plan to run night shifts may please owners and some customers needing faster delivery, yet upset residents and tired staff - showing conflict that must be managed explicitly.*

4 Forms of business ownership and sources of finance

As soon as businesses are set up and exchange goods and services with customers, the customers might wonder who - from a legal point of view - they are actually dealing with: the business itself or the people who founded and/or who run it. The answer to this question is of particular interest in the case of closing a contract, investing in the business or legal troubles: who owns the business? Who runs it? Who is the contracting party and who is to be sued in a litigation? Is the business a legal entity of its own or is the business completely identical with the owner(s) of the business?

The answer depends on the legal structure that has been chosen for the business. That choice has implications for ownership, management, liability, taxation and the range of financing options. Finance is important because businesses need money for their activities - at the start, to keep going, and possibly to expand. Although the different forms of legal structure as well as their denomination vary among countries, the basic forms are quite similar in many countries of the world.

4.1 Sole proprietorship / sole traders

A sole proprietorship is a business that is owned by one person who also manages and runs the business. It is easy to establish - especially for small businesses - because there are no financial requirements to start this kind of business. The business is not a legal entity of its own, so the profits are directly reported on the business owner's personal income tax statement. This means that the owner pays tax on the profits that are earned from the business.

Sole proprietorship (sole trader) A business owned by one person who also manages and runs it. The firm is not a separate legal entity.

The management of the business largely depends on the sole proprietor. On the one hand, he or she can make all management decisions and does not necessarily have to consider other opinions. On the other, continuity problems may occur if the sole proprietor wants to retire or suffers a (long-term) illness. If the sole proprietor needs support, he or she can hire personnel. However, it will always be his or her task to make the most important management decisions and take all the risk.

Financial funds of the business mainly depend on the financial capabilities of the sole proprietor. If he or she lacks financial funds, it will be very difficult to set up the business. Most sole proprietors invest their own savings in their business. If they need some extra money, they can try to get it from investors and/or from banks. The owner's investment and financial funds from other investors and creditors are external sources of finance.

Banks and other financial institutions offer different kinds of credit, short-term (duration less than a year) as well as long-term. The creditor, e.g. the bank, provides money for a certain time period which needs to be repaid according to the agreement, usually with a rate of return (the interest). Creditors usually ask for assets that can serve as collateral, especially for long-term credit. Especially long-term bank loans are often based on land and property as collateral (mortgage).

The most common forms of short-term credit are a bank overdraft and trade credit. A bank overdraft is a very flexible instrument, because once the bank account has been opened, the sole proprietor can withdraw money from the account when it is needed. Interest is only paid when the account is overdrawn. Trade credit is based on an agreement with the supplier. Usually a business does not have to pay all purchases immediately but is provided a trade credit period. All kinds of credit - short-term as well as long-term - are liabilities for the sole proprietor.

Unlimited liability The sole proprietor is liable for all debts and obligations, so his or her private assets are also at stake if the business fails.

As soon as the business starts operating and generates revenues, internal sources of finance can possibly be used. If revenues exceed expenses, the business makes a profit that can be retained and reinvested in the business (unless the profit is taken by the sole proprietor). Another option is the sale of assets that are not needed anymore. Internal sources of finance are very important to a business because no financial charges (e.g. interest) have to be paid for this sort of funds.

Worked example. *A neighbourhood bakery is set up by one baker who invests €12,000 of personal savings in ovens and fittings and opens a current account with a €3,000 overdraft facility. Flour and packaging are bought on 30-day trade credit. In year one the bakery earns €28,000 profit before drawings; the owner withdraws €20,000 for living costs and leaves €8,000 in the business. The €8,000 is an internal source of finance. If sales collapse and unpaid supplier invoices and the overdrawn account cannot be cleared from business cash, creditors can claim against the baker's private assets.*

4.2 Partnership

If two or more persons jointly found a business, this business is called a partnership. They need to set up a partnership agreement in order to settle the rights and responsibilities as well as the division of profits and losses. In a general partnership all partners have equal rights, liabilities and responsibilities. They can share the tasks and specialise. In difficult situations they can exchange their ideas and (maybe) make better decisions. In Austria, this form of business ownership is called "Offene Gesellschaft", or "OG" in short.

General partnership A partnership in which partners jointly own and manage the business and each partner has unlimited liability for the debts of the firm. In Austria this form is called Offene Gesellschaft (OG).

Two consultants who want to share tasks and risk with equal rights and responsibilities would typically choose such an Offene Gesellschaft. They will (have to) set up a partnership agreement in which they specify each partner's percentage of ownership, the division of profit and loss, terms of the partnership, rights and responsibilities, decision making and resolving disputes, and other details of the partnership.

In general, the financial aspects of partnerships are similar to those of sole proprietors. However, two (or more) partners should possibly be able to invest more savings and raise more financial funds than a sole proprietor. It also seems plausible that they can offer more (private) assets as collateral for getting a loan. Each of the partners is solely liable for all debts of the business (unlimited liability).

Limited partnership A partnership with at least one partner who is not involved in management and whose liability is limited to the amount contributed. In Austria this form is called Kommanditgesellschaft (KG).

Worked example. *Two consultants found a general partnership (OG). Partner A contributes €20,000 and Partner B €10,000; they agree to share profits 60:40 after salaries for work done. The firm borrows €40,000 from a bank for office renovation. When the firm cannot repay and business assets are exhausted, the bank may claim the full remaining debt from either partner - not only 60% from A and 40% from B. If instead they had formed a KG with an investor as limited partner contributing €25,000 and taking no management role, that investor's loss would normally be capped at €25,000,* *while the managing partners would still face unlimited liability.*

4.3 Corporations

Corporations are businesses that are legal entities of their own, which means that they have the same rights and obligations as people: legal persons can - just like natural persons - own land and property, hire people, close contracts, sue and be sued. The people who found the corporation and own a share of the business need not be involved in the management of the business, and the managers need not own a share of the business. Corporations are more difficult to set up, but the shareholders' liability is usually limited to the amount of money they invested when buying the shares.

Corporation An incorporated business with separate legal personality. Ownership is divided into shares; shareholders' liability is typically limited to the capital they invested.

So if shareholders only provide the money for share capital but are neither obliged nor entitled to manage the company, who does? The corporation is managed by the board of directors, persons who are elected by the shareholders to make all major business decisions and to represent the shareholders. The highest-ranking manager of this board is called the Chief Executive Officer (CEO). Other board members are the Chief Financial Officer (CFO), Chief Operating Officer (COO), Chief Information Officer (CIO) or Chief Marketing Officer (CMO).

Corporations usually have more options to raise financial funds than sole proprietors and partnerships. Their financial funds mainly comprise share capital as well as loans and credit. The capital of a corporation is divided into shares, which is why it is called share capital. If share capital equals 1,000,000 euros and is divided by 100,000 shares, each share (also called stock) represents 10 euros or 0.001 per cent of the share capital. Persons who buy shares become shareholders. Shares can be bought either at the time they are initially issued by the corporation or later on from some other shareholder who sells his shares. If all 100,000 shares are actually sold, the corporation gains 1,000,000 euros as share capital. This example shows that huge amounts of money can be raised from the sale of shares. Share capital is usually not redeemed by the company. It is long-term capital or even permanent capital.

A corporation's stock can but does not have to be listed on a stock market or stock exchange. A stock exchange is a financial market, regulated by the authorities, where shares and other securities (e.g. bonds) can be easily bought and sold by lots of people and businesses. The corporations that want their shares to be listed have to comply with certain rules and fulfil listing requirements. The shares are introduced on the stock exchange at a certain price (this introduction is also called initial public offering (IPO)) and then the prices are determined by demand and supply. According to the laws of supply and demand, usually the prices go up if demand for shares is high and vice versa.

There are a lot of reasons why demand for shares of a corporation can change: rising demand can be due to people's expectations that the business is doing well, that it will make profits in the (near) future, that it will increase market share and/or successfully introduce a new product. Demand for shares is also influenced by economic indicators like economic growth, interest rates and inflation. Usually there is higher demand for shares when the economy is thriving (because many people have money to invest), with comparatively higher rates of inflation (because share prices would also increase) and comparatively low interest rates (because high interest rates would make other investments more attractive). Please note that an increase in share prices after they have been issued does not have any additional financing effect for the issuing corporation. The beneficiaries of this increase are the shareholders only.

- **People invest in shares to support a business they believe in.**

- **They may seek dividends - parts of profit paid to shareholders (with no automatic obligation to pay every year).**
- **They hope for capital growth if share prices rise.**
- **Common stock typically includes voting rights at the stockholders' meeting.**
- **Shares represent a claim on real economic activity that may hold value better under inflation than some cash holdings.**

Among the biggest and most well-known stock markets of the world are the New York Stock Exchange (NYSE), Nasdaq Inc. in New York, London Stock Exchange (LSE), Tokyo Stock Exchange, Hong Kong Exchanges and Clearing, and Shanghai Stock Exchange. Apart from the LSE, the most important stock exchanges in Europe are the Deutsche Börse in Frankfurt, the NYSE Euronext (Europe), the SIX Swiss Exchange, Borsa Italiana in Milan, and the Spanish Exchange.

In some English-speaking countries, the public limited company (PLC) is one type of such a corporation. The term PLC is more commonly used in the United Kingdom and some Commonwealth countries. In the USA and other countries, you would rather find the denomination "Inc.", which stands for "Incorporated", or "Ltd" for "Limited". In German-speaking countries this form of legal structure for a business is called "Aktiengesellschaft" (abbreviated AG) as "Aktien" means "shares" and "Gesellschaft" stands for "corporation". The minimum capital requirement for founding an AG is 70,000 euros. Another type of corporation is the European Company (SE, société européenne). It is similar to an "Aktiengesellschaft" or PLC governed by Community law directly applicable in all - and only in - member states of the European Union.

Key figures on shares that are of interest for investors include share price and its development, market capitalisation or "market cap" (the total market value of a company's outstanding shares multiplied by the current market price), dividend yield (the dividend in relation to the share price), and the price-earnings ratio (P/E ratio) per share (a company's current share price relative to its earnings per share). A P/E ratio of about 16 means that an investor can expect to invest almost 16 euros in order to receive one euro of the company's earnings. A low P/E ratio usually indicates either that the shares are currently undervalued or that the company is thriving and earnings are high.

Apart from loans and credit that other businesses can also use, big corporations also have the option to issue bonds. Bonds can be regarded as a loan between investors as creditors and a corporation. The investors provide the corporation with a certain amount of money for a specific time period, based on an agreement when to repay the debt and how much interest to pay. Issuing bonds can be more attractive to corporations than borrowing from banks. The interest rate for bonds is often less than the interest rate for a bank loan. Borrowing large sums of money at low interest rates enables corporations to invest in long-term assets like plants, new infrastructure and other large-scale projects.

There are forms of private limited companies in many countries of the world. As a form of corporation, they are legal persons and the owners' liability is limited. However, shares usually are not sold to the general public at the stock exchange but are offered to the other owners of the company. The Limited Liability Company (LLC) in the USA, private company limited by shares in the UK, and the "Gesellschaft mit beschränkter Haftung", or GmbH in short, in Austria and Germany are examples of such private limited companies. The minimum capital for such companies is 35,000 euros in Austria - lower than for an AG, still a barrier compared with a partnership, and without the public flotation path of a listed AG.

Worked example. *A manufacturing corporation issues 200,000 shares at €25 each and raises €5,000,000 of share capital at IPO. Two years later the shares trade at €40 on the exchange. The* *market capitalisation is 200,000 × €40 = €8,000,000, but the corporation does not receive the €15 per-share price rise: that accrues to investors trading among themselves. Separately, the same firm issues a €2,000,000 bond at 4% interest to build a plant. Bondholders remain creditors, not owners.*

4.4 Summary: Overview of forms of business ownership

The various forms of business ownership can be summarised by differentiating between unincorporated and incorporated businesses. While unincorporated businesses are not legal entities of their own, incorporated businesses are legal persons.

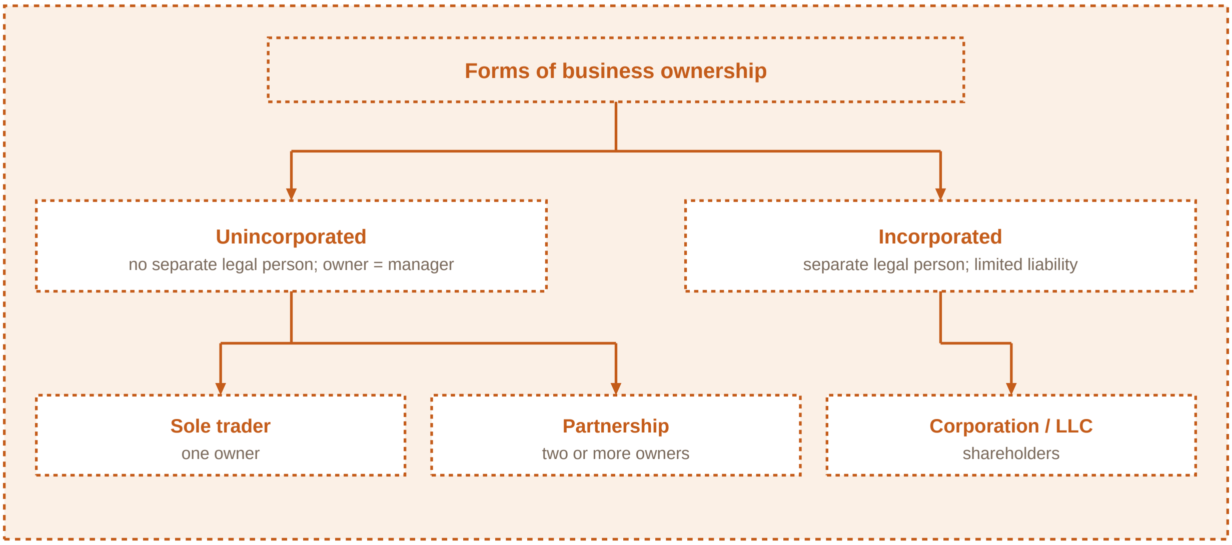

Figure 6. Forms of business ownership: unincorporated versus incorporated structures.

Among unincorporated businesses, owner(s) and manager(s) are typically the same person(s). With one owner, the structure is a sole trader / sole proprietorship. With more than one owner, the structure is a partnership. Among incorporated businesses, owners and managers need not be the same persons: shareholders provide share capital while directors run the company. This group includes corporations and limited liability companies.

Criterion	Sole trader	Partnership	Corporation
Legal personality	None - firm identical with owner	None - unincorporated	Separate legal person
Owners	One person	Two or more partners	Shareholders
Management	Owner manages	Partners manage (limited partners do not)	Board / directors; owners need not manage
Liability	Unlimited	Unlimited for general partners; limited for limited partners (to contribution)	Limited to capital invested in shares
Taxation of profits	Personal income tax of owner	Typically personal tax on partners' shares	Corporate framework (company and/or shareholders)
Continuity	Fragile if owner retires or falls ill	Depends on agreement and partner succession	Stronger - legal person continues
Typical finance	Owner savings, overdraft, trade credit, bank loans	Partners' capital, loans, overdraft, trade credit	Share capital, loans, bonds (larger firms)
Setup difficulty / minimum capital	Easiest; usually none	Agreement required; usually none	Harder; AG ≈ €70,000, GmbH ≈ €35,000 (Austria)

Table 5. Comparison of sole trader, partnership and corporation

Worked example. *Three founders compare options for a logistics start-up needing €80,000 of vehicles. As a sole trader, one founder would own everything, face unlimited liability, and fund mostly from personal savings and a bank loan. As an OG, all three manage and share unlimited liability while pooling capital. As a GmbH (meeting minimum capital rules), all three take shares, liability is limited to invested capital, and they appoint managing directors - who may or may not be shareholders.*

4.5 Overview of sources of finance

The table below gives an overview of the most commonly used sources of finance. The balance sheet reveals the sources of finance that a business has used (see Chapter 6, Accounting). Large businesses are obliged to draw up such a balance sheet.

Equity finance Funds that belong to the owners of the business (for example share capital and retained earnings).

Debt finance Borrowed funds that create liabilities: bank overdrafts, trade credit, bank loans, bonds and similar instruments.

Equity finance (internal and external)	Debt finance (external)
Funds provided by investors (external)	Short-term credit: e.g. bank overdraft, trade credit, short-term (bank) loans
Share capital (external)	Long-term credit: e.g. bank loan, loan provided by owners, bonds
Retained earnings (internal)	

Table 6. Sources of finance

Equity finance may be internal or external. External equity includes funds provided by investors and share capital raised from owners who buy newly issued shares. Internal equity centres on retained earnings - profits kept in the business rather than distributed. Debt finance is external: creditors stand outside ownership. A bank overdraft lets a firm draw beyond the credit balance on its current account up to an agreed limit; interest is charged on the overdrawn amount. Trade credit is supplier-granted delay in paying for purchases. Bank loans provide a fixed sum for an agreed term with scheduled repayment. Bonds package large-scale borrowing from multiple investors with contractual interest.

Worked example. *A retailer finances a shop refit as follows: €50,000 retained earnings (internal equity), €30,000 new capital from a silent investor (external equity), a €20,000 three-year bank loan (long-term debt), a €5,000 overdraft for seasonal stock (short-term debt), and €8,000 of inventory purchased on 45-day trade credit (short-term debt). Selling an unused delivery van for €4,000 adds internal financing without interest.*

4.6 The choice of the source of finance

If a business has several sources of finance available that it can choose from, it will most probably make its decision based on costs, the intended use of the financial funds and its current financial situation.

- **Costs comprise interest payments for loans and credit or administration costs for issuing shares or bonds.**
- **If the financial funds are used for capital expenditures (e.g. for buying assets that will be used over some or even many years), long-term finance is required. Revenue expenditures (e.g. buying material that is used for production) can safely be financed by short-term sources.**
- **A business that already has a high proportion of loan capital (also called a "high geared" business) might have difficulties obtaining more credit. Lenders might be reluctant to offer more funds, only at a higher price (a higher interest rate) and/or if collateral can be offered. Such a business should rather try to use internal sources of finance and/or find investors or business partners who are willing to provide funds. As loans must be repaid, a high proportion of loan capital can be a burden (risk of insolvency).**

High gearing A high proportion of loan capital relative to equity, which raises insolvency risk and may make further credit harder or more expensive to obtain.

Worked example. *A workshop wants a €120,000 CNC machine with a ten-year useful life and €15,000 of steel for next month's orders. Financing the machine with a revolving overdraft would mismatch duration: the bank could demand repayment long before the machine pays for itself. A ten-year loan or lease matches better. The steel can sensibly sit on trade credit or a short-term facility. If the firm's loans already equal 75% of capital employed (high gearing), adding another large* *loan may be denied or priced punitively; retaining profit or admitting an equity partner may be the feasible path.*

5 Marketing

Producing a good or providing a service only makes sense if there are enough people who are interested in the good or service and ask for it. It is not only businesses that seek to make a profit that need sufficient demand for their goods and services. Non-profit organisations can also only exist over a longer period of time if their goods and services meet a demand. Therefore, "having a market" and knowing the market is highly important to all kinds of businesses.

Marketing activities intend to achieve this goal. They also intend to tailor the products to customers' needs and wishes and to continuously improve the product. Marketing is much more than good advertising, although this is the most visible part of marketing. According to the American Marketing Association, marketing is "the activity, set of institutions, and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large". Either way, marketing is about creating value for the customer.

5.1 What a product is

Product (marketing) In marketing terminology, a product is every good and/or service that can be exchanged in order to fulfil the wishes and needs of customers.

Customers can be either other businesses or consumers (private households). Producer products are goods and services that are sold from one business to another (business-to-business, B2B), while consumer products are sold to consumers or private households (business-to-consumer, B2C). If a consultancy buys a printer for its office, it is a producer product. If the exact same printer is not used in the office but in a private home, it is a consumer product. A tutoring start-up may offer support services mainly to private households (B2C), but also to other businesses (B2B).

Benefits for the customer may be the functionalities of the product but also some additional features, the image that it conveys and the joy it gives to its user. A mobile phone creates value, for example, because you can reach your family and friends from wherever you are (so it creates value through functionalities), and you can also receive emails, messages and surf online (additional features). If you buy the latest model of a certain mobile phone with lots of additional functionalities, everyone thinks you are an expert in the field and that makes you happy.

Worked example. *A firm sells industrial cooling units to factories (producer products, B2B) and compact air-conditioners to households (consumer products, B2C). It also sells annual maintenance contracts - pure services, still products in marketing terms. A technician who buys a unit for the company workshop is in a B2B exchange; the same model purchased for a living room is B2C.*

5.2 Objectives of marketing

In order to fulfil customers' wishes and needs, a business has to analyse the market(s) it wants to operate in and be clear about its marketing objectives. They may include but are not limited to:

- **Customer satisfaction: If customers are not satisfied with the product, they will not buy it (again). Satisfied customers often become loyal customers who will possibly buy the product again.**
- **Creating a unique selling proposition (USP): If a product has a USP, it is or is considered to be - in one way or another - different from other similar products. This is also called product differentiation and can be based on a certain characteristic of the product or on the way it is promoted and perceived by customers. Building a brand supports creating a USP.**

- **Gaining and maintaining market share: Businesses will be interested in having a certain share of the market - it indicates their relative importance compared to their competitors' positions.**
- **Maintaining or increasing sales: Sales are important because businesses need the revenues to cover production costs and to make a profit.**
- **Profitability: Usually the higher the sales figures are, the more profit a business will make (with certain limitations). Profits reimburse owners for invested money and can be retained and reinvested.**

Unique selling proposition (USP) A distinctive benefit or perceived difference that sets a product apart from similar offerings and supports differentiation and customer preference.

Worked example. *A regional bicycle brand sets objectives for the year: raise average customer satisfaction scores, build a USP around lasting frames and free annual service, lift absolute market share in the urban-commuter segment from 8% to 10%, grow unit sales by 12%, and keep operating margin above 9%. Promotion emphasises the service USP rather than pure price cuts.*

5.3 Product orientation versus market orientation

Over the past decades, many businesses have moved from a product-oriented approach towards a market-oriented approach. While a product-oriented business focuses on the product and its specifications first and later thinks about how to sell it, the market-oriented business will first analyse customers' needs and wants and then tailor their products to fulfil these requirements. Both businesses may have some of the marketing objectives mentioned above (they may even be identical), but the product-oriented business mainly relies on the quality of the features of the product to be successful in the market.

Product orientation An approach that prioritises production and product features first, then seeks to sell what has been made.

Market orientation An approach that starts from customers' needs and wants and designs offerings to match them.

The market-oriented approach has the advantage of anticipating changing needs and wants and of responding to these market changes earlier than the product-oriented competitors. It largely depends on the product itself and the number of competitors which of the two approaches is more suitable for a business. However, the market and the expectations of customers should not be neglected.

Customer relationship management (CRM) aims at creating a long-term relationship with customers. Their data is kept to mail or email them newsletters, coupons and product information so that they will come back and buy again. Using personal data has become a very important topic and the sensitive use of personal data is indispensable. However, by using personal accounts and loyalty cards, customers often give up their anonymity willingly in order to receive discounts and use other special offers. So many businesses collect data on their customers and their buying behaviours and can tailor their offers to their customers' personal preferences.

Worked example. *A kitchen-appliance maker once designed ever-more complex mixers and then funded heavy advertising (product orientation). After stagnating sales, it interviewed home cooks, found demand for simpler, quieter machines that are easy to clean, and redesigned the line accordingly (market orientation). Loyalty-card data later showed which accessories were bought together, enabling CRM emails with relevant add-ons while offering clear opt-out choices.*

5.4 The need for more responsibility and sustainability

It is often claimed that businesses not only focus on what customers need or want, but that they also create wishes and needs by continuously developing new products and advertising them, sometimes even in unethical ways. The fact is that many people spend more money than they can afford to or at least more than they initially intended to spend. They buy things that they do not really need and will not use (or just for a very short period of time). Therefore, it seems desirable that both businesses and consumers become more aware of the importance of sustainable production and consumption and realise the potential risks of consuming too much. Both businesses and consumers need to act in a more responsible and sustainable way.

A repair-and-reuse business hopes to contribute to sustainability: devices should be used longer and not immediately replaced if a problem occurs, but repaired and reused. There are similar tendencies in the market for clothes. More and more people no longer want to buy cheap clothes that are thrown away after a few months. They prefer high-quality clothes that they can give to their friends and might get some other good clothes in return. Renting high-quality clothes is another trend. Instead of buying (cheap) clothes for a particular event you rent clothes of higher quality and return them afterwards.

Worked example. *An electronics retailer replaces "upgrade every year" campaigns with a repair desk, certified refurbished devices and transparent adverts about battery health. A fashion label launches a rental line for formal wear and accepts trade-ins for recycling fibre. Households that previously bought three cheap outfits per event now rent one high-quality set and return it.*

5.5 Market research

Market research provides information about existing and prospective customers of a business, the (potential) buyers of its product(s), about the competition, and the industry in general. This information, the research data, can be based on two sources.

Primary information is gained by conducting an empirical study or having data collected by a market research institute. This study can be tailored to the needs and interests of the business. A business might be interested to know who their customers are, which products they buy, what they think about these products, if they also buy similar products of other businesses. A bakery, for example, would love to know more about prospective customers: which products do they need or want? How much are they willing to pay? How much and which kind of support do they need? And do men and women, older and younger customers differ in their answers?

Primary market research Original data collection designed for the commissioning business's questions (surveys, interviews, observation, experiments).

But conducting a study can be very costly: administering hundreds or thousands of questionnaires, conducting personal or telephone interviews, setting up an online survey, analysing and interpreting the data costs a lot. Especially small businesses often cannot afford this sort of market research.

Secondary information is based on existing research that has already been done by somebody else (and maybe for some other purpose). Government agencies, trade and/or industry associations or other businesses often conduct research that is available free of charge and might also serve the purpose of learning more about the market. However, in most cases, the information will be very general and not tailored to the specific interests of a business.

Secondary market research Use of existing data collected by others (statistics, published reports, industry studies), typically cheaper but less specific.

Customer analysis will often be an important element of market research activities. In order to learn more about (potential) customers, a business will try to learn more about:

- **Who its current and potential customers are: If they are consumers, the business operates in a B2C market, otherwise in a B2B market. Also, the buyer and the user of a product might not be the same persons.**
- **What the customers do with the products: The more a business knows about preferred use, the more it can develop and improve the product.**
- **Where the customers buy the products: This helps identify preferred channels of distribution.**
- **When the customers buy the products: This can identify seasonal fluctuations and support price differentiation over the year.**
- **Why customers choose a product or prefer another: Motives and preferences matter for product development and market share.**

Market research can also focus on measuring the market. Market size (or market volume) is the total sales of a product of all the businesses in the market. Market size can be expressed as a value (e.g. in euros) or as a quantity (number of pieces sold). Market share refers to the proportion (percentage) of a certain market that is held by a business or its products or brands. It is calculated by dividing the sales of a business by the amount of total sales in the market.

Absolute market share: Absolute market share = sales volume of one business (or one brand) / market volume

Relative market share: Relative market share = market share of a business (or a brand) / largest competitor's market share

Absolute market share is important information for the business itself, but also for (potential) investors. However, this figure does not tell a lot about the other competitors in the market. Relative market share shows how a business (or one of its brands) is doing in terms of its largest competitor. Based on the assumptions that there are potential customers in the market that could be won, the market potential exceeds market volume, and the sales potential of a business exceeds its current sales volume.

Worked example. *A firm's sales in a defined market are €150,000; total market volume is €1,000,000. Absolute market share = 150,000 / 1,000,000 = 0.15 = 15%. The largest competitor holds 30% absolute share. Relative market share = 15% / 30% = 0.5. If market potential is estimated at €1,200,000, about €200,000 of demand remains unrealised industry-wide.*

5.6 Market segmentation and targeting strategies

Market research may reveal that some characteristics (like age, gender, education level, income level, behavioural patterns, interests etc.) may be relevant for describing the (potential) customers of a business. These characteristics can possibly be used to differentiate between relatively homogeneous subgroups of customers (market segmentation). Segmentation makes sense for a business if the segments are measurable (size, purchasing power etc.) and profitable, accessible by communication and distribution channels, and durable (not changing too quickly). If such segments can be found, the business selects the segment(s) that it wants to target with its products (targeting). The next step is to position the products so that it becomes clear which product meets the demands of the targeted subgroups. Positioning is the process by which marketers try to create an image or identity in the minds of their target market(s).

Base	Examples of variables	Typical use
Geographic	City, region, country; urban versus rural	Local reach, logistics, climate or language differences
Demographic	Age, gender, education, income, employment status	Purchasing power and life-stage offers
Psychographic	Lifestyle, values, attitudes (e.g. environmental concern)	Message tone and brand meaning
Behavioural	Usage intensity, occasions, loyalty, benefits sought	Pricing, service intensity, loyalty programmes

Table 7. Common segmentation variables

Market segmentation Dividing a market into relatively homogeneous subgroups of customers who share relevant characteristics.

A business applies mass marketing if it ignores the different segments of a market and offers the same product to all customers. The product is promoted to all segments of a market in almost the same way. Mass marketing is often applied for products like pens and pencils, soaps, personal hygiene products and detergents because they are used by so many people regardless of their personal characteristics. Mass marketing allows a business to produce a relatively large number of the same product and sell it to a relatively large market. On the one hand, this enables the business to gain economies of scale: if a large number of identical products is produced, some costs will not increase accordingly (in a directly proportional way) but can be divided by a larger number of products. This results in reduced costs per unit due to an increased total output of a product. On the other hand, mass marketing is also inflexible and makes it more difficult to react to particular changes in some target markets.

Segment marketing means offering different products to one or more segments (some segmentation). Businesses focus their (limited) resources on offering products that are tailored to the needs and wants of their customers in the segments that they know best (strategic fit). Niche marketing, however, focuses on particular segments of a market. Different products are offered to subgroups within segments (more segmentation). Many small businesses rather employ niche marketing because they would not be able to produce the quantity of products for a mass market. Some businesses just target selected segments of a market because they have specialised in a certain field. If they are good at it, they can be market leaders regardless of their size.

Target market A group of people or businesses towards whom a business markets its goods, services or ideas with a strategy designed to satisfy their specific needs and preferences.

Worked example. *A refurbished-laptop seller defines a geographic focus on one metro region, demographics of employed adults and retirees, psychographics favouring lower-cost and lower-waste devices, and behaviour of occasional rather than heavy users. It targets small offices and home offices with limited IT budgets (segment) and within that specialises in buyers needing setup support (niche). Positioning stresses "ready-to-use, supported, affordable and durable" rather than "latest model".*

5.7 The marketing mix

A marketing mix is a harmonised blend of several marketing tools that best meets the needs and wants of the customers of the targeted market. It consists of the four elements product, price, place and promotion - the four Ps.

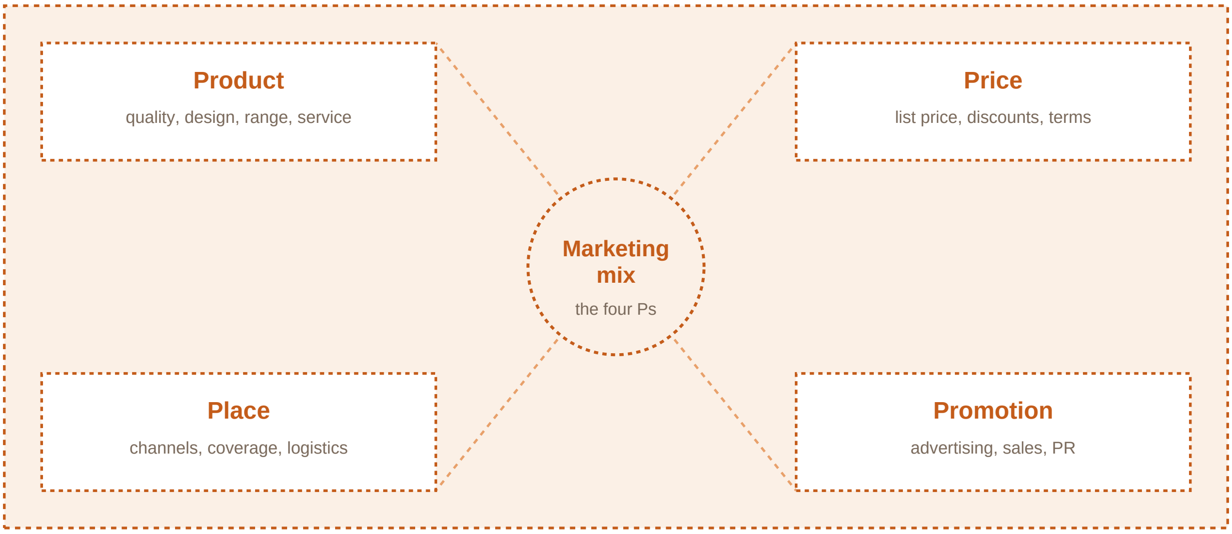

Figure 7. The four elements of the marketing mix: product, price, place and promotion.

It is the basic idea of the marketing mix to provide a good at an affordable price at a convenient place and communicate a certain message in order to promote the sale of the product. The first P, "Product", refers to all goods and services that are offered by a business. This "P" is always at the heart of marketing and the most important decision of a business. Most businesses offer not just one product, but a range of products. Products that are very similar (like several slightly different laptop computers) represent a product line. A business can produce just one product line or different product lines.

Brands are created to support product differentiation. A brand consists of a name or a (few) word(s) and/or a symbol and/or a sign. It distinguishes a product or the business from other products and businesses and is intended to build a USP and brand recognition as well as brand loyalty. Brands also serve as a guarantee of stable quality and of a certain level of quality that is maintained all over the world. Brands can render trust and safety of choice for a customer.

According to the product-mix strategy of a business it can specialise in one sort of product (by offering just one product line) or diversify its production (by offering different product lines, increasing product mix width) or even both. As consumers' needs and wants change over time, products need to be altered from time to time. Minor changes (like different packaging, colours, etc.) are called a relaunch. If a relaunch is not enough to keep customers satisfied, then major changes are necessary or the product has to be eliminated from the product line altogether. Businesses can either add new products to an existing product line (which is a line extension) or they can add a new product line (which is a mix extension). Products can also be altered over time or eliminated from the product range if a relaunch does not seem promising.

The product life cycle represents a theoretical model of several stages over the life span of a product that differ in sales volume and in profit. Before a product is introduced to the market, there are no sales, so no money is flowing into the business. However, there are costs from product development, so the introduction phase starts with a loss. After the product launch, the product begins to sell. But costs might still be higher than sales because the business needs to spend a lot on promoting the product and introductory prices might be low to attract customers. In this phase, the product can be a so-called "question mark". According to the Boston Consulting Group matrix, a product with low relative market share in a market that is rapidly growing is a "question mark". Towards the end of the introduction phase, revenues exceed costs and a (small) profit is made.

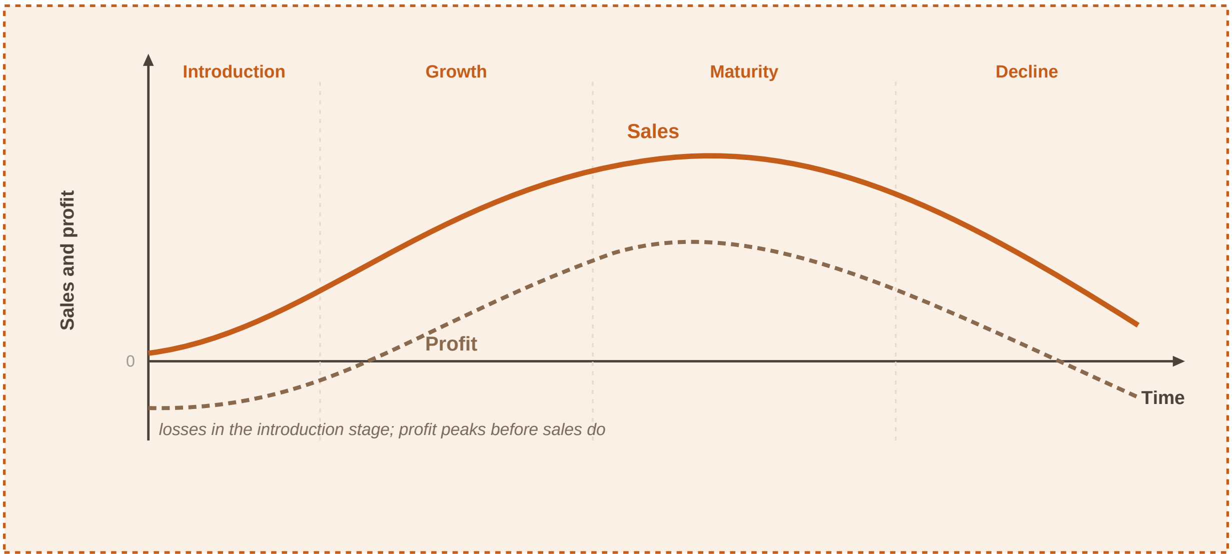

Figure 8. The product life cycle: introduction, growth, maturity and decline.

During the growth period, sales increase more rapidly than costs (average costs might even fall due to increased output and economies of scale). As market share increases in a market that is still growing, the product may become a "star". Stars are very valuable to businesses because of their strong market position. In order to maintain this position, businesses need to invest money in promotion and in production facilities. Usually, the business finally makes a profit in the growth stage that peaks in the maturity stage. As soon as the market is not growing that fast anymore, but market share is still high, the product becomes a "cash cow". Businesses usually do not invest so much money anymore because market growth has become low. So revenues are (still) high but expenses decrease. However, profits will - sooner or later - tend to fall again due to increased competition. The product more and more becomes a "poor dog" as it nears the decline stage. Sales fall and so do profits.

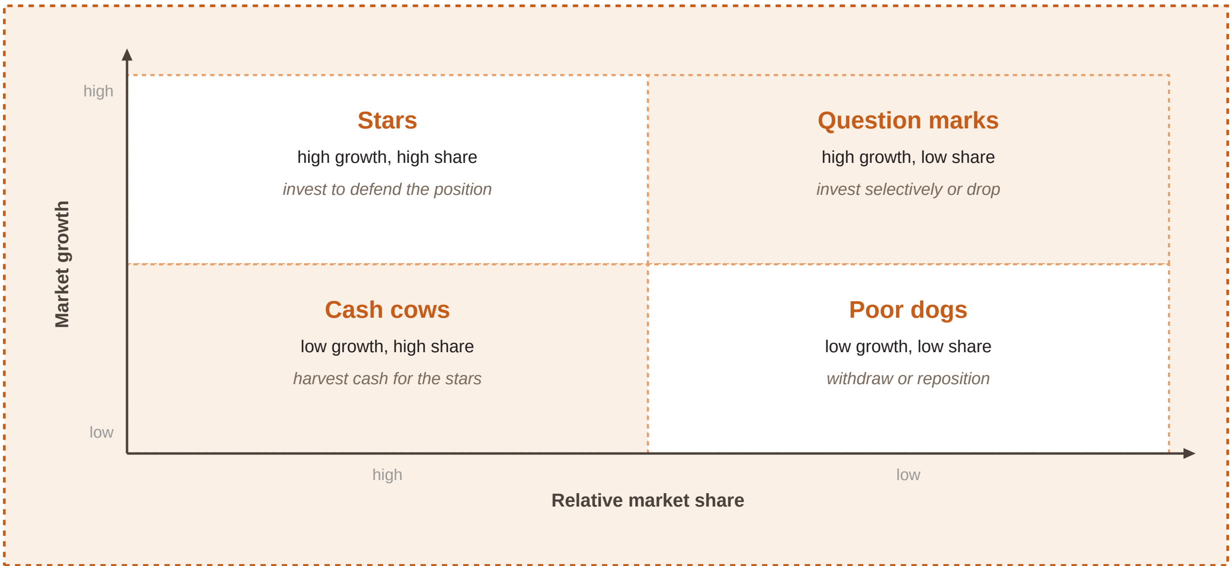

Figure 9. The Boston Consulting Group matrix: stars, cash cows, question marks and dogs.

The product portfolio of a business should be balanced - a blend of stars, cash cows, question marks and not too many poor dogs. Revenues from the cash cows can be used to support the stars (because they are the future cash cows) and invest in question marks that the business believes in. The product life cycle varies enormously in duration across product categories. The maturity stage can take months or years - some products like detergents, toothpaste or perfumes seem to have an indefinite maturity phase. Other products ("fads") have a very short life cycle and have to be withdrawn after less than a year.

In order to prevent sales from declining, a business may decide to use a strategy to extend the life cycle of a product (an "extension strategy"). Changing the product mix and entering new markets are the most important extension strategies, as can be seen in Ansoff's product-market matrix. Market penetration (existing product, existing market) is the safest of the four options. Product development (new product, existing market) is slightly riskier. With market development (existing product, new market), an existing product is put into an entirely new market. Diversification (new product and new market) is the riskiest of the four options, but can also reduce risk in the long run by spreading across lines and segments.

The second P, "Price", is also crucial and difficult to determine. Once a price has been set, it is very hard to change the pricing strategy because that would confuse customers. The lowest possible price is not always the safest choice to appeal to customers. The main influencing factors are costs (because costs need to be covered and - possibly - some profit gained), the prices charged by the competition, and demand (because it is also important to consider how much customers are willing to pay).

In the long run, all businesses need to cover their total cost if they want to survive. Total costs comprise fixed costs and variable costs. Variable costs increase directly as output rises. Fixed costs are independent of the output, like the rent or insurance for the office. As soon as revenues exceed total cost, the business "breaks even". The output that needs to be sold in order to "break even" is also called the "break-even point".

Contribution and break-even: Contribution per unit = selling price – variable cost per unit
Break-even units ≈ fixed costs / contribution per unit

In variable-cost-plus pricing, a markup is added to the variable cost of producing a good (or service). If a firm adds a 50% markup and the variable cost of one refurbished laptop is 480 euros, then the selling price would be 480 + 240 = 720 euros (excl. VAT). The absolute lower limit for the price of a good in the short run is variable cost: as soon as the price exceeds variable costs, the excess contributes to covering fixed costs. Therefore, this amount is also called "contribution". In competitive markets, businesses usually employ distribution pricing when setting prices for new products or for contract bidding: they add a certain markup to variable costs.

Before changing prices it makes sense to consider how customers might react. How demand is affected depends on the price elasticity of demand. Demand is "elastic" if the change in demand (in per cent) exceeds the price adjustment (in per cent). Demand is "inelastic" if the quantity response is smaller than the price change in percentage terms. Few substitutes and necessities tend toward inelastic demand; luxury and easily substituted goods toward elastic demand.

Place (distribution) is easy to underestimate, yet a strong product fails if customers cannot find it where they expect it. Depending on the product, firms use distributors, wholesalers (who buy and resell to businesses), retailers (who buy and resell to consumers), agents or brokers (common in B2B), online selling, telemarketing, and formats such as vending machines or kiosks. Channel partners bridge factory and final buyer: logistics, local information and advertising support.

Promotion covers activities that inform potential customers and the wider public about the business, its products and their benefits. The promotion mix includes advertising on TV and radio, online and social media, newspapers and magazines, billboards, buses and trains; direct mailing; personal selling through salespeople; sales promotions; sponsorship of events, people or organisations; and public relations that build a favourable image and stakeholder relationships.

Small budgets may emphasise websites, local ads, social presence, leaflets and word-of-mouth from satisfied customers. Whatever the tools, promotion must match the target segment and stay consistent with product quality cues, price positioning and place.

Worked example - contribution and break-even. *Variable cost per unit €480, selling price €720, fixed costs €130,000. Contribution = 720 – 480 = €240. Break-even units ≈ 130,000 / 240 = 541.67, so from the 542nd unit the firm moves into profit (simplified model). If one customer offers to buy ten units for €6,000 total (€600 each), the firm still covers variable costs (€4,800) and earns €1,200 contribution toward fixed costs - worth accepting unless the same units can surely be sold at the full €720 elsewhere.*

6 Accounting - keeping record of business transactions

During their first year in business, firms buy and sell goods, support customers, hire assistants and deal with paperwork and bookkeeping. This means that a lot of business transactions like purchases, sales, borrowing and lending money take place and businesses need to keep accurate records of these business transactions. Bookkeepers are responsible for recording these transactions that are always verified by a document (e.g. an invoice or a receipt). The aim of recording all transactions is to generate information on the financial status of the business. Statements that provide this information are called accounts.

This chapter explains the three main components of the financial statement - the balance sheet, the income statement (profit and loss account), and the cash flow statement - and how to read and analyse them.

6.1 What a balance sheet is

When a business starts, it uses equipment it already has as well as new purchases. Computers, office equipment and cars - these things are all assets, "things" that the business owns and that are used for the business. Owners make a list with all the assets of their business and their values in euros (on the left side of the list) and compare it with the amount of money that had to be borrowed from the bank (on the right side of the list) to pay for some of their purchases.

Please note that there are two types of computers that may appear in such a list: computers that are used in the office and are not for sale, and other computers that were bought to be resold (inventory). This difference is important because they are different types of assets. Only part of the assets may be financed by using money from a bank loan. The rest may be financed by the owners' own resources. Therefore, the total amount of the bank loan (which is a liability, i.e. money that is owed to someone else) is lower than the total amount of the assets. The difference between these two positions is the amount that the owners were able to finance themselves; it is called owner's equity. It is the proportion of the assets that was NOT financed by debt. If we add this position to the list, the list becomes a balance sheet.

A balance sheet comprises a company's assets and reveals how they were financed. The amount of assets equals the amount of liabilities plus owner's equity.

Balance-sheet identity: Assets = liabilities + owner's equity

Assets		Equity and liabilities	
Office equipment	25,000	Owner's equity	24,000
Van	8,000	Bank loan	25,000
Inventory	12,500		
Cash and bank	3,500		
Total assets	**49,000**	**Total equity and liabilities**	**49,000**

assets = equity + liabilities (both sides always balance)

Figure 10. The balance sheet: assets financed by liabilities and owner's equity.

Worked example - constructing a balance sheet. *A newly founded consultancy holds the following resources on 1 January: office equipment and computers used in the office €25,000; a delivery van €8,000; computers held in stock for resale (inventory) €12,500; cash and bank deposits €3,500. A bank loan of €25,000 financed part of these purchases; the rest was financed by the owners.*

Total assets = 25,000 + 8,000 + 12,500 + 3,500 = €49,000.
Liabilities (bank loan) = €25,000.
Owner's equity = assets – liabilities = 49,000 – 25,000 = €24,000.

Assets	€	Liabilities and equity	€
Office equipment (incl. computers)	25,000	Owner's equity (capital)	24,000
Van	8,000	Bank loan (liabilities)	25,000
Inventory	12,500		
Cash and bank deposit	3,500		
Total assets	49,000	Total liabilities and equity	49,000

Table 8. Simplified opening balance sheet (€)

- **All assets were funded either through equity or liabilities.**
- **All funds are somehow bound or invested in the business (even money held as cash or in a bank account was either funded through equity or liabilities).**
- **Any increase in assets must be financed by an increase in either liabilities or equity (capital).**

Please note that if the firm buys computer software and pays cash, the total amount of the assets remains the same (one asset - software - increases, but another one - cash - decreases by the exact same amount, so it is just an "asset swap"). But if they buy the software on credit, then the amount of assets increases and liabilities also increase by the exact same amount. Accordingly, the balance sheet total increases.

Assets are usually listed in the balance sheet in a certain order. They comprise fixed assets (or non-current assets) that normally have a lifespan of more than one year and are intended to be used in the company for a longer time period than one year. Usually they cannot be turned into cash so easily. Current assets have higher liquidity. They are usually not used longer than a year because they are used up, spent in production or sold.

Non-current (fixed) assets Assets with a useful life of more than one year that are intended for longer-term use in the business (e.g. property, plant, machinery, office equipment).

Current assets Assets with higher liquidity that are normally used up, sold or converted into cash within one year (e.g. inventory, receivables, cash).

There are also intangible assets like trademarks, patents and copyrights. Although they cannot be seen or touched they are also of value for the business like tangible assets are. Liabilities are debts and obligations that are owed to other persons, businesses or banks and need to be repaid at a certain point in time and/or over a certain period of time. These funds are usually provided by banks or suppliers. Money owed to suppliers is also called trade credit or trade payables. Current liabilities are debts or obligations that are due within one year. Accordingly, non-current liabilities have a duration of more than one year.

Equity / owner's equity is the difference of assets and liabilities. It is an indicator of the wealth of the company. A high equity ratio (= equity / total capital) indicates that the respective portion of assets was financed by the company's own resources (for example with funds provided by the owners or shareholders of the company). It is important for a business to have sufficient equity because: 1) equity usually does not have to be repaid, 2) it helps a business to be relatively independent from its creditors and 3) if the company has a loss and there is a higher amount of equity, the company will not be over-indebted.

Equity ratio: Equity ratio = equity / total capital (total assets)

Determining the value of assets is an important but also tricky task. While it might not be difficult to determine the exact amount of a bank loan, it is more difficult to tell what a building is worth many years after it was built. Businesses usually want to present themselves in the best way possible when setting up their balance sheet. However, they need to stick to the legal rules and regulations that aim to ensure that information given in balance sheets can be trusted.

Worked example - asset swap versus financed purchase. *Start from total assets €49,000, liabilities €25,000, equity €24,000.*

Case A - buy €2,000 of software for cash:
Software +2,000; cash –2,000. Total assets still €49,000. Liabilities and equity unchanged.

Case B - buy the same €2,000 of software on credit:
Software +2,000; trade payables +2,000. Total assets €51,000; liabilities €27,000; equity still €24,000. Identity: 51,000 = 27,000 + 24,000.

6.2 Other components of the financial statement of a business

The balance sheet is not the only component of the financial statement of a business. The financial statement of the business aims to show the performance of the business over a certain period of time, but the balance sheet only shows the assets, liabilities and equity at a certain point in time. It does not contain information on total sales (turnover), nor does it reveal the cost of producing the goods and services or other expenses. Therefore, the financial statement consists of the balance sheet, the income statement (profit and loss account) and the cash flow statement, which gives an insight into the inflow and outflow of cash.

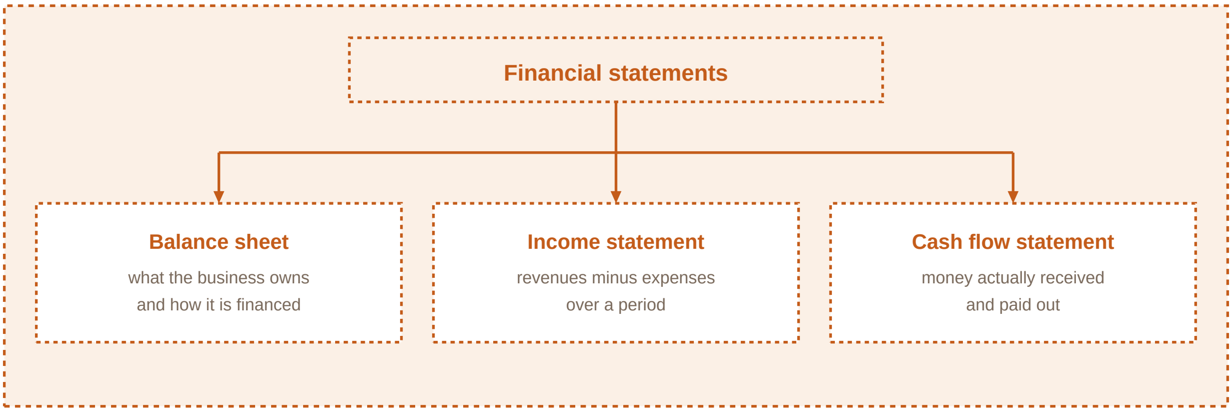

Figure 11. The main components of a financial statement: balance sheet, income statement and cash flow statement.

Income statement (profit and loss account) A period statement that summarises all revenues, costs and expenses over a time span (for example a year) to determine whether the business made a profit or a loss.

Revenues are the income (usually cash or accounts receivable) that is generated from selling goods or services to customers. Costs consist of resources that are consumed in order to produce the goods or provide the services. If the revenues exceed the costs and expenses, the company has a profit. If the company's costs and expenses are higher than the revenue, it suffers a loss.

> ***Worked example - simple profit.*** *In one financial year a service firm records sales revenue of €400,000 and costs and expenses (materials, wages, rent, energy, depreciation and other expenses) of €310,000.*
> *Profit = 400,000 – 310,000 = €90,000.*
> *That profit increases equity by €90,000 if it is retained in the business.*

Depreciation Recognition that the value of fixed assets decreases as they are used up over time. Depreciation is an expense in the income statement, but unlike wages or energy it does not cause an actual cash payment in the period when it is charged.

Fixed assets are used again and again over a longer period of time. A delivery van, for example, will be used for at least five more years. A new computer will be used for three years. During that time, the value of these assets decreases; it is "used up". Without depreciation, the values of assets that are shown in the accounts would be inaccurate. The value of the assets in the balance sheet would be overstated and the financial information given would be flawed.

Straight-line depreciation (per year): Annual depreciation = depreciable cost / expected useful life
(Equal charge each year; land is not depreciated.)

> ***Worked example - straight-line depreciation.*** *A firm buys a computer for €2,100 and expects to use it for three years (expected useful life = 3 years), with no residual value for simplicity.*
> *Annual depreciation = 2,100 / 3 = €700 per year.*
>
> *Book (carrying) values at year-end:*
> *After year 1: 2,100 – 700 = €1,400.*
> *After year 2: 1,400 – 700 = €700.*
> *After year 3: 700 – 700 = €0.*
>
> *Each year the income statement includes a €700 depreciation expense that reduces profit, while the*

cash outlay of €2,100 occurred when the computer was purchased.

At the end of a trading year, owners draw up another balance sheet to see how assets and capital structure have changed. Due to depreciation, fixed assets have a lower value than one year before. New equipment may have been bought, inventory levels adjust, some customers still owe money (accounts receivables), and the firm may still owe suppliers (trade credit or accounts payables). Part of a bank loan may have been repaid. Equity has increased significantly when the firm made a profit. To learn more about the performance of a business (profit or loss), you can either look at the income statement or at the development of (owner's) equity. It is important to note that any profit increases and any loss decreases equity.

Assets	€	Liabilities and equity	€
Office equipment (incl. computers)	42,000	Owner's equity	114,000
Van	40,000	Bank loan	12,500
Car	6,000	Trade credit / payables	15,000
Inventory	18,000		
Receivables	12,500		
Cash and bank deposit	23,000		
Total assets	141,500	Total liabilities and equity	141,500

Table 9. Simplified year-end balance sheet after a profitable year (€)

Cash flow statement A statement that reveals the flows of cash into and out of the business during a period, distinguishing operating, investing and financing activities. A positive cash flow is not identical with a profit.

This is important information because cash is needed for the operating activities of the business and for investing. To learn more about a cash flow's origin, a cash flow statement is prepared that differentiates between changes in cash positions in operations (the core activities of a business), investments and financing activities.

- **Cash flow from operations shows how well a business generates cash with its core business - usually the most important part and ideally positive.**
- **Cash flow from investments shows how cash is spent for long-term investments or generated by the sale of such assets. A negative figure often simply indicates that the business has invested.**
- **Financing activities refer to cash flowing in from investors or creditors and flowing out for paying interest, dividends or repaying debt.**

Worked example - total cash change versus profit. *Opening cash (including bank deposits) is €3,500; closing cash is €23,000.*
Total cash increase = 23,000 – 3,500 = €19,500.

Suppose the same year reports a profit of €90,000. Profit and the cash increase differ because profit includes non-cash charges such as depreciation and accruals (sales on credit, unpaid expenses), while the cash figure only counts actual cash movements. A firm can be profitable yet short of cash, or can raise cash through borrowing without earning a matching profit.

Worked example - classifying cash flows. *Classify these cash movements for a manufacturing firm in one year:*
• Customers pay €120,000 of trade receivables -> operating inflow.
• Raw materials paid in cash €45,000 -> operating outflow.
• New machine bought for €80,000 cash -> investing outflow.
• Bank loan repayment €15,000 -> financing outflow.
• Dividend paid to owners €10,000 -> financing outflow.

Net operating cash (simplified) = 120,000 – 45,000 = €75,000 inflow.
Investing cash = –€80,000. Financing cash = –€25,000.
Approximate net cash change from these items = 75,000 – 80,000 – 25,000 = –€30,000.

6.3 What can be learnt from reading a balance sheet and an income statement

Balance sheets and income statements should always be read with some caution. A lot depends on the valuation of assets and businesses are not completely free to determine the values of their assets. Depreciation also decreases the value of assets in the balance sheet that might actually have a higher value. Nevertheless, a lot can be learnt by taking a closer look at the balance sheet and the income statement of a business. It is of particular interest to see how some positions have developed over time as well as in comparison to other competitors in the market.

This allows us to answer questions like:

- **Which assets does the business have - is there a higher percentage of current or non-current assets and is this typical of that type of business?**
- **How has equity developed and how have the assets been financed? Is there a higher percentage of long-term or short-term liabilities?**
- **Is there a balance between non-current assets and long-term financial resources?**
- **How have revenues developed over the past year? Have costs, especially costs of sale, developed accordingly?**
- **How have profits (or losses) developed?**

The first three questions can be answered by analysing the balance sheet. Questions about revenues, costs and profits require a look at the statement of profit and loss. Cost of sales (COS) (or cost of goods sold (COGS)) are the costs that are directly tied to the production of the products like the cost of labour (that is directly linked to production), materials and manufacturing overhead. Administration costs, the cost of shipping to customers or the cost of sales personnel is NOT included in COGS. Gross profit reveals the earnings of a business after deducting the direct costs of producing the goods (without operating expenses). Earnings before interest, taxes, depreciation and amortisation (EBITDA) allow analysis of operating performance while including operating expenses differently; by deducting depreciation and amortisation, the next metric is earnings before interest and taxes (EBIT).

In accounting, expenditures and expenses are not the same. While expenditures are payments that are either made to purchase (non-current as well as current) assets, expenses are costs that have expired or were "used up" in order to produce the goods or provide the services that were sold. For example, COGS are considered to be expenses, and so are other expenses such as salaries, marketing costs, interest, insurance, rent, and so on.

Worked example - reading structure and performance (euros in thousands). *Suppose a manufacturing company reports at year-end: non-current assets 944; current assets 586; total assets 1,530. Equity 711; non-current liabilities 515; current liabilities 304; total equity and liabilities 1,530.*

Share of non-current assets = 944 / 1,530 ≈ 61.7%. A high fixed-asset share is typical of manufacturing that needs plant and machinery.
Equity ratio = 711 / 1,530 ≈ 46.5% - almost half of assets financed by equity.
Long-term finance available = equity 711 + non-current liabilities 515 = 1,226, which exceeds non-current assets 944. So long-term assets are covered by long-term resources.

Income statement: revenue rises from 815 to 992 (+177, about +21.7%), while cost of sales rises from 760 to 830 (+70, about +9.2%). Gross profit therefore rises sharply. Revenues grew faster than cost of sales - a favourable operating development.

6.4 Use of these accounts - types of accounting

These accounts are needed for a number of stakeholders who are interested in the financial situation of a business. Some of them are internal users like the owners of the business, the managers and the employees. They are all interested to learn if the business is thriving. Managers need to make decisions based on financial information (like "Do we need to cut costs in marketing?") and owners will want to know if their investment is worth the risk (like "How much profit do I get in return for investing and risking my money?"). There are also external users of financial information like tax authorities, suppliers, competitors, investors and the media.

For some companies it is mandatory to have their accounts checked for authenticity by an independent firm of accountants, an auditing company. This task is called auditing. The result of the auditing process can be read in the annual report.

Managerial accounting Accounting focused on providing information for the management of the business so managers can decide where to cut costs, how to calculate prices, and how to allocate resources.

Financial accounting Accounting aimed at producing information - such as the balance sheet and the income statement - that is of interest both inside the firm and to external decision makers (tax authorities, banks, investors).

Of course, managers are also interested in financial accounting, something that is also of interest for decision makers outside a business like tax authorities or banks. Information gained from financial accounting like the balance sheet and the income statement can be found in the annual reports for example. Many large businesses also publish selected information on their financials on their websites.

Worked example - who needs which information. *A bakery considers raising the price of a specialty bread. Management accounting estimates cost per loaf (€1.80 materials and labour), contribution after variable costs, and effect on monthly volume if the retail price rises from €3.20 to €3.50.*

Separately, the bank financing a new oven asks for the latest balance sheet and income statement (financial accounting): total equity €90,000, bank loan €40,000, last year's profit €25,000. Tax authorities use the same financial accounts to assess taxable profit.

6.5 Analysis of financial statements

The information given in the balance sheet and in the income statement can be used to calculate figures and ratios in order to learn more, e.g. about the liquidity (or the solvency) of a business, profitability (which refers to the relationship of profit to the capital employed or to the turnover), financial efficiency (which deals with the question how effectively a business has employed its resources) and financial structure (which can be evaluated on the basis of equity ratio and debt ratio).

Please note: Since these figures and ratios can vary a lot from one industry to another, some comparisons would not be very meaningful. Comparisons are meaningful for different businesses within the same industry or sector or for one single business over time (comparisons of several different financial years). There might also be different ways to calculate some of these ratios, so it is always important to know how ratios were calculated if you want to make comparisons.

Theme	Ratio	Formula
Structure	Equity ratio	equity / total assets
Structure	Debt ratio	total debt / total assets
Liquidity	Working capital	current assets – current liabilities
Liquidity	Current ratio	current assets / current liabilities
Liquidity	Acid-test ratio	(current assets – inventory) / current liabilities
Profitability	Return on equity	profit before interest and tax / equity
Profitability	Return on capital employed	profit before interest and tax / capital employed
Profitability	Capital employed (one definition)	equity + non-current liabilities (or assets – current liabilities)
Efficiency	Asset turnover	turnover / average assets
Efficiency	Inventory (stock) turnover	cost of sales / average inventory
Shares	Earnings per share	profit for ordinary shareholders / shares outstanding
Shares	Price-earnings ratio	share price / earnings per share
Shares	Dividend yield	dividend per share / share price
Shares	Market capitalisation	shares outstanding × market price

Table 10. Selected ratio formulas used in financial analysis

The term liquidity refers to the ability of a business to pay its bills and repay its debts on time. One approach to evaluate liquidity is to calculate the working capital of a business. Working capital (or circulating capital) indicates whether or not a business is able to pay its day-to-day bills such as electricity, rent and wages and buy components for its production. Therefore, the focus is on short-term assets and liabilities.

Working capital: Working capital = current assets – current liabilities

The calculation of working capital is based on the idea that current assets of a business are relatively liquid (meaning that they can be easily turned into cash). If all the current assets are turned into cash to repay current liabilities, working capital is the amount that is left over after all current debts have been paid. Working capital should be positive, meaning that current assets should be higher than current liabilities. If they are not, this would indicate that part of the non-current assets was financed by short-term liabilities, which might cause problems when these liabilities need to be repaid.

Working capital problems and cash flow problems are interlinked but not identical. Cash flow problems can be mitigated by borrowing money using a short-term bank loan or using overdraft facilities and trade credit. Although this brings cash into the business, it will increase current liabilities and reduce working capital. The most effective way to improve both working capital and cash flow is to increase equity or borrow more long-term credit.

Liquidity ratios: Current ratio = current assets / current liabilities
Acid test ratio = (current assets – inventory) / current liabilities

As current assets should exceed current liabilities, the current ratio should be greater than 1, ideally even between 1.5 and 2. It can be argued that inventories should not be taken into consideration for this calculation. There are many reasons why inventories could possibly not be turned into cash. This is why a modified calculation, the so-called acid test ratio, does not consider inventories and allows a stricter evaluation of liquidity.

Worked example - liquidity ratios (euros in thousands). *Current assets 586; inventory 136; current liabilities 304.*

Working capital = 586 – 304 = 282.
Current ratio = 586 / 304 ≈ 1.93 (above 1, near the common 1.5-2 range).
Acid-test ratio = (586 – 136) / 304 = 450 / 304 ≈ 1.48 (still above 1).

Interpretation: short-term resources exceed short-term claims even after excluding inventory, so liquidity looks adequate on these measures - subject to industry comparison.

If revenues exceed costs, the business makes a profit. This does not necessarily mean that the business is sufficiently profitable because the amount of profit might be minuscule in relation to the money that has been invested (and the risk that has been taken). Hence, profitability refers to a business's profit in relation to an indicator of the size of the business (total assets or total average assets, total equity, turnover).

Return ratios: ROE = profit before tax and interest (EBIT) / equity
ROCE = profit before tax and interest (EBIT) / capital employed
Capital employed ≈ equity + non-current liabilities (or assets – current liabilities)

Worked example - return ratios (euros in thousands). *Operating profit before interest and tax 90; equity 711; total assets 1,530; current liabilities 304.*

Return on equity = 90 / 711 ≈ 12.7%.
Capital employed = 1,530 – 304 = 1,226.
Return on capital employed = 90 / 1,226 ≈ 7.3%.

If average capital employed were 1,168, ROCE = 90 / 1,168 ≈ 7.7%. These figures are most useful when compared with peers in the same industry and with the firm's own history.

Financial efficiency refers to the question how efficiently a business has employed its resources. Two of the ratios that can be calculated to evaluate financial efficiency are asset turnover and inventory (stock) turnover. Asset turnover aims to indicate how much turnover was generated by every euro invested in total average assets or - alternatively - in net assets. Inventory turnover shows how many times a business has sold or used up and replaced inventory. High inventory turnover is important because it indicates that goods sell well and do not remain in stock for a long time.

Efficiency ratios: Asset turnover = turnover / average assets
Inventory (stock) turnover = cost of sales / average inventory

Worked example - turnover ratios (euros in thousands). *Turnover 992; assets beginning 1,437; assets ending 1,530.*
Average assets = (1,437 + 1,530) / 2 = 1,483.5.
Asset turnover = 992 / 1,483.5 ≈ 0.67 (about €0.67 of sales per €1 of average assets).

Cost of sales 830; average inventory 122.5.
Inventory turnover = 830 / 122.5 ≈ 6.8 times per year -> stock renewed roughly every 365 / 6.8 ≈ 54 days.

When analysing listed companies, investors also use share-related figures published with financial results. Earnings per share divide profit attributable to ordinary shareholders by the number of shares outstanding. The price-earnings ratio divides the current share price by earnings per share. Dividend yield expresses the dividend relative to the share price. Market capitalisation (shares outstanding × market price) gauges size but is not necessarily a meaningful metric for the fundamental value of a company, because there are many reasons why share price might be particularly high or low. After shares already trade on an exchange, a rise in market price does not by itself bring new cash into the company - only sellers benefit; new equity cash comes from issuing new shares.

Worked example - earnings per share and price-earnings ratio. *Profit attributable to ordinary shareholders €5,000,000; shares outstanding 2,000,000; market price €20.*

Earnings per share = 5,000,000 / 2,000,000 = €2.50.
Price-earnings ratio = 20 / 2.50 = 8.
An investor pays €8 of price per €1 of current earnings.

If the firm buys back 200,000 shares and profit stays €5,000,000, shares outstanding become 1,800,000 and earnings per share rise to ≈ €2.78 - higher per-share earnings without any change in total profit.

End of Full Course theory

Continue with practice tasks for the chapter you have just studied.